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MILITARY SPECIFICATION

CHASSIS, TRAILER, SINGLE- AND MULTI-AXLE

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers various sizes of single- and multi-axle cart-type trailer chassis having a low-silhouette frame, suitable for skid-mounted operating equipment. The trailer chassis are compatible for towing behind trucks, both commercial and military design and for being hauled behind construction and industrial machinery (CIM) equipment. These trailer chassis are compatible with a variety of operating machinery equipment, work tool sets, and shop sets which supplement and assist other equipment on earthmoving, construction, and materials handling jobs.

1.2 Classification. Trailer chassis shall be of the following sizes, as specified (see 6.2). The size of trailer shall be determined by the minimum DOT Gross Vehicle Weight Rating (GVWR) as follows:

Size 1, CCE	-	Min. 1400 lb. GVWR, for light truck	-	MIL-C-52437/1
Size 3, CCE	-	Min. 3000 lb. GVWR, for light truck	-	MIL-C-52437/2
Size 4, CCE	-	Min. 4000 lb. GVWR, for medium truck	-	MIL-C-52437/3
Size 6, CCE	-	Min. 6500 lb. GVWR, for medium truck	-	MIL-C-52437/4
Size 8, CCE	-	Min. 8000 lb. GVWR, for medium truck	-	MIL-C-52437/5
Size 9, MD	-	Ref. 8700 lb. GVW, for medium military design 2-1/2 & 5 ton M-Series trucks	-	MIL-C-52437/6
Size 11, CCE	-	Min. 11,000 lb. GVWR, for heavy truck.	-	MIL-C-52437/7
Size 14, CCE	-	Min. 14,000 lb. GVWR, for heavy truck.	-	MIL-C-52437/8

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2330

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Size 32, CCE - Min. 32,000 lb. GVWR, for heavy truck.
Size XX, CCE - As specified by the procuring activity (see 3.1.1 and 6.2, T-1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
GGG-J-51	- Jack, Mechanical Hand Operated (Screw, Scissors, Reel, Planer, and Jack Bar).
KKK-T-2107	- Trucks, and Truck Tractors, Commercial, Diesel or Gasoline Driven, 14,000 to 21,000 Pound GVW, 4 x 2.
KKK-T-2108	- Trucks and Truck Tractors, Commercial Diesel or Gasoline Driven, 24,000 to 32,000 Pounds GVW, 4 x 2.
KKK-T-2109	- Trucks and Truck Tractors, Commercial Diesel or Gasoline Driven, 34,500 to 64,000 Pounds GVW, 6 x 4.
KKK-T-2110	- Truck and Truck Tractors, Commercial Gasoline and Diesel Engine Driven, 16,000 to 34,000 Pounds GVW, 4 x 4.
KKK-T-2111	- Truck and Truck Tractors, Commercial Diesel Engine Driven, 36,000 to 60,000 Pounds GVW, 6 x 6.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.

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MIL-T-740	- Truck, 5 Ton, 6 x 6, Military Design, M39 Series.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-13486	- Cable, Special Purpose, Electrical, Low-Tension, Heavy Duty Single-Conductor and Multiconductor.
MIL-C-13486/12	- Cable, Special Purpose, Electrical, Low-Tension Heavy Duty Multiple Conductors Unshielded.
MIL-S-40626	- Sign Kit, Vehicle Class.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone Automotive All Weather Operational and Preservative.
MIL-T-46780	- Truck, 2-1/2 Ton, 6 x 6, Military Design, M44A2 Multi-Fue1 Engine Series.

(See Supplement 1 for list of associated specifications.)

STANDARDS

FEDERAL

FED-STD-292	- Truck, Light Commercial, Gasoline/Diesel Fueled, Four Wheel Driven (4 x 4, 3000 to 11,000 Pounds GVW).
FED-STD-307	- Trucks, Light Commerical, Gasoline/Diesel Fueled, Two Wheel Driven (4 x 2, 3000 to 11,000 Pounds GVW).

MILITARY

MIL-STD-105	- Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobile, Trucks, Truck-Tractors and Trailer Dollies, Preservation and Packing of.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1179	- Lamp, Reflectors and Associated Signaling Equipment for Military Vehicles.
MIL-STD-1180	- Safety Standards for Military Ground Vehicles.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

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MS51331	- Guard, Splash, Wheel.
MS75020	- Connector, Plug, Electrical – 12 Contact, Intervehicular, 24 Volt, Waterproof.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13212E3630	- Chassis, Trailer, Trailing Arm, 2 Wheel, 2-1/2 Ton.

DEPARTMENT OF LABOR (DOL)

29CFR 1910 - Occupational Safety and Health Standards (OSHA).
29CFR 1926 - Construction Safety Standards.

(Application for copies should be addressed to the local DOL Office or the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

49CFR 393 - DOT Federal Motor Carrier Safety Regulations (FMCSR).
49CFR 567 and 568 - Certification.
49CFR 571 - DOT Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, 710 North Capital Street, NW, Washington, DC 20402.)

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting office.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z35.1 - Industrial Accident Prevention Signs, Specification for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN PLYWOOD ASSOCIATION (APA)

US Product Standard PS 1.
Plywood Design Specification Manual X215.

(Application for copies should be addressed to the American Plywood Association, 1119 A Street, Tacoma, WA 98401.)

AMERICAN SOCIETY FOR METALS (ASM)

ASM Source Book on Brazing and Brazing Technology.
Iron Casting Handbook.

(Application for copies should be addressed to the American Society for Metals, Metals Park, OH 44073.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 242	-	Steel, Structural, High Strength Low Alloy.
A 449	-	Bolt & Studs, Quenched & Tempered Steel.
A 710	-	Low-Carbon Age-Hardening Nickel-Copper-Chromium Molybdenum-Columbium and Nickel-Copper-Columbium Alloy Steels.
E 84	-	Surface Burning Characteristics of Building Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF AGRICULTURAL ENGINEERS (ASAE)

ASAE R276.

(Applications for copies should be addressed to the American Society of Agricultural Engineers, P.O. Box 229, St. Joseph, MI 49085.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standards.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN TRUCKING ASSOCIATION, INC. (ATA)

The Maintenance Council (TMC-ATA):

ATA Vehicle Maintenance Reporting Standards.
TMC-ATA Recommended Practice Manual (Formerly RCCC).
TMC-ATA Mechanic's Personal Tool Inventory.

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(Application for copies should be addressed to the Maintenance Council - ATA, 1616 P Street, NW, Washington, DC 20036.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS Welding Handbook.
AWS D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society Inc., 2501 North West 7th Street, Miami, FL 33125.)

AMERICAN WOOD PRESERVATION

AWPA Standards.

(Application for copies should be addressed to the American Wood Preservation Association, 7735 Old Georgetown Road, Bethesda, MD 20014.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open Top Cars.
M-931 - Highway Trailers, all Types, for TOFC Service.

(Application for copies should be addressed to the Association of American Railroads, 1920 L. Street, NW, Washington, DC 20036.)

ANTI-FRICTION BEARING MANUFACTURES ASSOCIATION INC. (AFBMA)

AFBMA Standards for Ball and Roller Bearings and Balls

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers Association, Inc., 1235 Jefferson Davis Highway, Suite 704, Arlington, VA 22202.)

FORGING INDUSTRY ASSOCIATION (FIA)

The Forging Industry Handbook.

(Application for copies should be addressed to the Forging Industry Association, Dept. R, 1121 Illumination Building, Cleveland, OH 44113.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC. (FMS)

FMS Automotive Data Book.

(Application for copies may be obtained from commercial brake lining contractors or should be addressed to the Friction Materials Standards Institute, Inc., East 210, Route 4, Paramus, NJ 07652.)

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INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards
OMFS Recommendations.

(Application for copies should be addressed to Industrial Fasteners Institute, 1505 East Ohio Bldg., 1717 East 9th Street, Cleveland, OH 44114.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS, INC. (NAME)

Metal Name Plate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Metal Name Plate Manufacturers, Inc., 1000 Vermont Avenue, NW, Washington, DC 20005.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Agent, American Trucking Association, Inc., 1616 P Street, NW, Washington, DC 20036.)

NATIONAL WHEEL AND RIM ASSOCIATION (NWRA)

NWRA Catalog of Truck Wheels.
NWRA Cross Reference List.
NWRA Load Ratings for Truck Wheels.

(Application for copies should be addressed to the National Wheel and Rim Association, 4837 Victor Street, Jacksonville, FL 32207.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California, Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC Vol. 2 - Systems and Specifications

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(Applications for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Ave., Pittsburgh, PA 15213.)

TIRE & RIM ASSOCIATION, INC. (T&RA)

T&RA Year Book.

(Application for copies should be addressed to the Tire & Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC.

TTMA - Recommended Practices.
TTMA - Technical Bulletins.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 1020 Princess Street, Alexandria, VA 22314.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification and its specification sheets shall take precedence.

3. REQUIREMENTS

3.1 Description. The trailer chassis shall be as specified herein and on the applicable size specification sheets (see 1.2). The trailer chassis shall consist of a round tow eye, draw bar structure, support leg (landing jack), equipment bed frame with standardized crossmembers, rear suspension, running gear, brake system, and electrical system. The equipment bed shall accept both skid mounted end items and equipment which can be mounted on a skid to mate with the standardized trailer framing. The equipment bed shall be compatible with various sets and assemblies of engineer type operating equipment (see 6.1) with the exception of the size 9 military design trailer chassis, all other trailers shall be of the commercial construction equipment (CCE) types. These CCE trailer chassis shall be equipped with all components, furnished as standard and optional equipment, whether specified herein or not, that are necessary to enable the trailer chassis to function safely, reliably, and efficiently in engineer and construction job applications. Size 9 trailer chassis shall be of the military design type as required on the drawings and herein.

3.1.1 Specification sheets. The individual specifications of each size of trailer chassis shall be covered in the individual specification sheets. Chassis shall be furnished with standard components. When and as specified (see 6.2, T-1) the chassis shall be furnished with the optionally required components as specified by the contracting activity.

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3.1.2 Military Design. The size 9 military design (MD) trailer chassis shall be in accordance with Top Assembly, TA13212E3630, except where indicated herein and on the specification sheet for size 9 (see MIL-C-52437/6). Unless otherwise specified herein the trailer chassis shall be equipped with Dash -1 springs. When specified (see 6.2, T-2) furnish Dash -2 through Dash -6 springs for lighter equipment.

3.1.2.1 MD drawings. The drawings for the MD trailer chassis forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.1.3 Weights and dimensions. The trailer chassis shall be in accordance with table I of each applicable specification sheet and as summarized on the following table I, herein. The curb weight of the trailer chassis shall include the weight of the chassis complete with all accessories and components. The gross vehicle weight ratings (GWR), in accordance with DOT 49 CFR 567 and TTMA RP No. 39 (for highway speeds), shall be the curb weight of the chassis plus the weight of the payloads (equipment or component and body). Established components and vehicular ratings shall not be raised to conform to this specification. The physical characteristics of the trailer chassis shall be as specified in table I of the applicable specification sheets, when the trailers are on level surface, when the inflation of the tires are as recommended by the Tire and Rim Association for GVWR operation at highway speeds, and when the suspension systems are in the normal height position. When and as specified (see 6.2, T-3), the optional running gear and suspension shall be furnished as specified.

3.1.4 Standard commercial product. The CCE trailer chassis shall, as a minimum, be in accordance with the requirements of this specification and shall be in the line of the manufacturer’s standard commercial product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product shall be included in the machine being furnished. Standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements of manufacturer's catalogs, or brochures, and represents the latest production model. The capacities and rating of the chassis and any of its components shall not be raised to comply with this specification. The equipment shall be designed and constructed to facilitate field maintenance. All equipment of the same classification, furnished under a specific contract shall be physically and functionally identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations shall be obtained in advance from the contracting officer.

3.2 First article. The contractor shall furnish a trailer chassis for examination and test within the time frame specified (see 6.2) to prove that his production methods and choice of design

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TABLE I. Summary Chassis Characteristics*

			Size 1	Size 3	Size 4	Size 6	Size 8	Size 9	Size 11	Size 14	Size 32
GVWR Per DOT & VIN Plate.	Min.	Lb.	1400	3000	4000	6500	8000	8700	11,000	14,000	32,000
Equip Payload Allow.	Min.	Lb.	1100	2000	2800	5000	6000	5000	7800	11,000	23,000
Overall Length OAL	Max.	In.	148	148	190	216	222	175	222	330	330
Overall Height OAH -	-	In.	Minimum Practical								
Overall Width OAW	Max.	In.	70	80	96(80)	96	96	96	96	96	96
Bed Length	Min.	In.	80	96	120	144	144	117	144	244	228
Bed Height	Max.	In.	24	26	30	30	32	21	36	36	40
No. Axles, Std	Min.	-	1-ea.	1-ea.	1-ea.	1-ea.	1-ea.	1-ea.	2-ea.	3-ea.	2-ea.
Tire Size, Std.	Min.	-	4.80-12	7.00-16	7.50-16	9.50-16.5	12-16.5	9.00-20	12-16.5	9.5-16.5	10.00-15

*For reference only. See table I in each size specification sheet for contractual, detailed requirements.

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detail will produce equipment that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests, as specified (see 6.2).

3.3 Dissimilar metals. All dissimilar metals throughout the trailer chassis shall be insulated from one another to preclude galvanic or electrolytic action. Insulation shall be by a coat of epoxy, chromate, or alkaline-resistant, mylar barrier, electrical tapes, alumastic compound, or other effective means that prevent direct contact. Dissimilar metals are defined in MIL-STD-889 and MIL-T-704.

3.4 Material. Material shall be as specified herein and, for size 9 chassis as shown on the drawings. Materials not specified shall be selected by the contractor as being compatible with the intended use in accordance with the engineering standards of the trailer and construction equipment industry and as set forth in the following industry standardization publications:

Society of Automotive Engineers (SAE).
Truck Trailer Manufacturers Association (TTMA).
The Maintenance Council - ATA (TMC-ATA).
Tire and Rim Association (TRA).
Friction Materials Standards Institute (FSMI).
Industrial Fasteners Institute (IFI).
American Welding Society (AWS).
Forging Idustry Association (FIA).
American Society for Testing and Materials (ASTM)
American National Standards Institute (ANSI).

Rubber products furnished shall be not more than two (2) years from date of cure of the rubber to date of acceptance of the trailer chassis by the Governement. Threaded parts shall be in accordance with SAE J370, SAE J371, and Industrial Fasteners Institute Standards.

3.4.1 Material design criteria. The stresses in the load carrying members of the equipment, both as a plant jacked up for operation and as a trailer chassis prepared for road transport, shall not exceed 50 percent of the materials yield point (YP) strength under static conditions. A calculated stress analysis of the trailer chassis of the plant units shall be furnished. This analysis shall include calculations and diagrams for the following:

a. The gross vehicle weight rating (GVWR) load distribution diagram of each chassis, blocked and on wheels.
b. The vertical shear diagram for the length of each chassis, blocked and on wheels.
c. The bending moment diagram for the length of each chassis, blocked and on wheels.

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3.4.2 Recovered materials. All equipment, components, and materials incorporated in the trailer chassis furnished by the contractor shall be new and fabricated using materials produced from recovered material to the maximum extent possible without jeopardizing the trailer chassis intended use (see 6.1). The term "recovered materials" shall mean materials which have been collected or recovered from solid waste and reprocessed to become a quality source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or the furnishing of rebuilt products is allowed under this specification.

3.4.3 Material deterioration prevention. The materials used in fabrication of the trailer shall be either inherently corrosion resistant and weatherproof or protected from the deteriorating effects of exposure to the elements. The contractor shall have a written material and material finishing method, utilizing state-of-the-art technology, which identifies the specific material, material finish, or material treatment supplied on components and assemblies subject to weathering to provide protection against various forms of material deterioration in the trailer's operating and storage environments specified herein.

3.5 Maintainability. Components of the trailer chassis shall be accessible for servicing, repair, and replacement. The adjustment and replacement of components shall be accomplished with mimimum disturbance to adjacent components. Where incorrect installation of an item could cause malfunction of that item or of the system in which it is installed, an unsymmetrical mounting means shall be provided or the mating parts shall be doweled or matchmarked. Ease of maintenance provisions shall incorporate features insuring operating clearances in accordance with SAE J925, and facilitate maintenance and service operations. The trailer chassis shall be maintainable using only those common mechanic's tools which are listed in the "Mechanic's Personel Tool Inventory" as defined and published by Study Group S-5, The Maintenance Council - ATA. The trailer chassis Maintainability Index and Repairability Index, in accordance with SAE J817, shall be computed and included in the manual. The manuals for the chassis shall conform to SAE J920.

3.6 Transportability. The loaded trailer chassis at GVWR shall withstand impact forces encountered in vehicular, rail, marine, and air transport as specified herein without damage or permanent deformation.

3.6.1 Road transport. The trailer chassis shall comply with the over the road transport requirements in 3.7.2 and 3.7.3. When secured and loaded on a highway semitrailer, the trailer chassis shall be acceptable as cargo for road transport in accordance with the ATA National Motor Freight Classification Rules.

3.6.2 Rail transport. The trailer chassis shall withstand all forces encountered in rail transport service without damage or permanent deformation. The trailer chassis shall be loaded on a flatcar to comply with the requirements of the "Rules Governing the Loading of Commodities on Open Top Cars" of the Association of American Railroads. When specified (see 6.2, T-4), the chassis shall be subjected to a railroad humping test as specified in 4.6.2.8. Except for coupling to a trailer train fifth wheel coupler stand, which is incompatible with the tow eye, the trailer chassis shall conform to the applicable portions of AAR specification M-931.

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3.6.3 Marine transport. The trailer chassis shall withstand all forces encountered in overseas transport by marine ships as both inside the hull cargo and on-deck outside cargo.

3.6.4 Air transport. Unless otherwise specified herein, the trailer chassis and equipment shall be air transportable in C-5A aircraft as shown in figure 1 without requiring sectionalization. When and as specified (see 6.2, T-5), the equipment shall be sectionalized, as necessary, and the trailer chassis and its equipment shall be air transportable in C-130 aircraft as shown in figure 2. Sectionalization of each equipment loaded trailer chassis shall be accomplished in not more than 4 hours and sectionalization shall be done without requiring cutting or welding. All sectionalized segments shall be equipped with their own lifting and tiedown attachments.

3.6.5 Lifting and tie down attachments.

3.6.5.1 Lifting attachments. Each trailer chassis and, if applicable, each sectionalization component shall be provided with lifting attachments conforming to MIL-STD-209. The attachments shall enable the equipment to be lifted in its normal traveling or operating position. The attachments shall not result in damage or permanent deformation of the slings or of the exterior of the equipment. No spreader bars are permitted. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.6.5.2.

3.6.5.2 Tiedown attachments. Each trailer chassis and, if applicable, each sectionalization component shall be provided with integral attachment conforming to MIL-STD-209.

3.7 Performance characteristics. The trailer chassis shall meet the transport performance characteristics specified herein under the following conditions:

a. With the trailer lubricated and serviced (see 3.18 and 3.19).
b. With the specified equipment payload or at GVWR.
c. When towed by trucks or self-propelled equipment.

With the operating equipment installed and the trailer chassis jacked to level operating position, the trailer chassis shall withstand plant operation for extended time periods of at least a year.

3.7.1 Towing compatibility. The trailer chassis shall be compatible for coupling, towing, maneuvering, spotting, and under all conditions specified herein, when towed by trucks, construction equipment, and materials handling equipment, to include those specified below:

- Trucks having 20-to 25-inch pintle height for towing trailers having up to 75 percent of the truck's VIN plate GVWR; 12 volt electrical – no air for braking:

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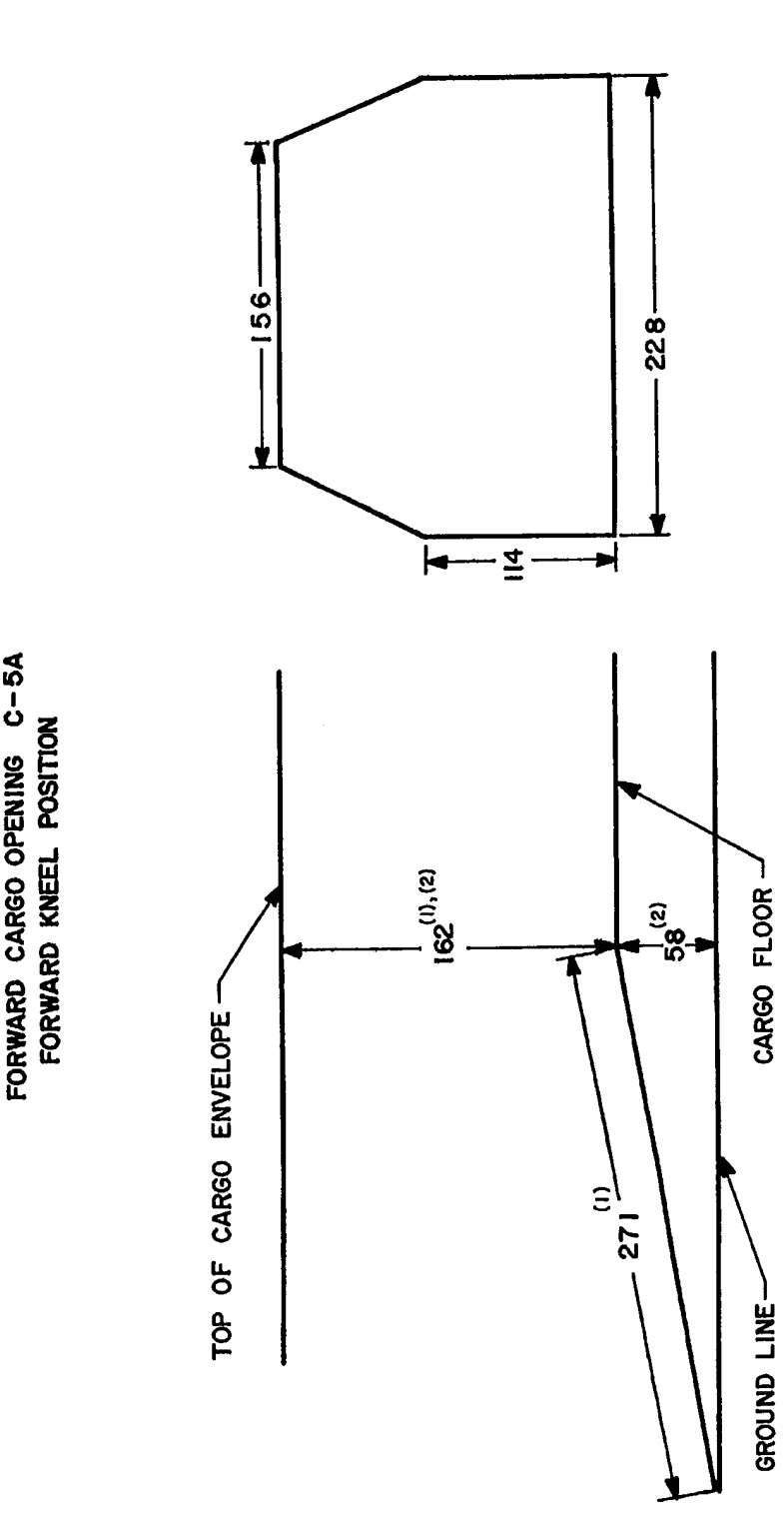


FIGURE 1. C-5A cargo envelope.

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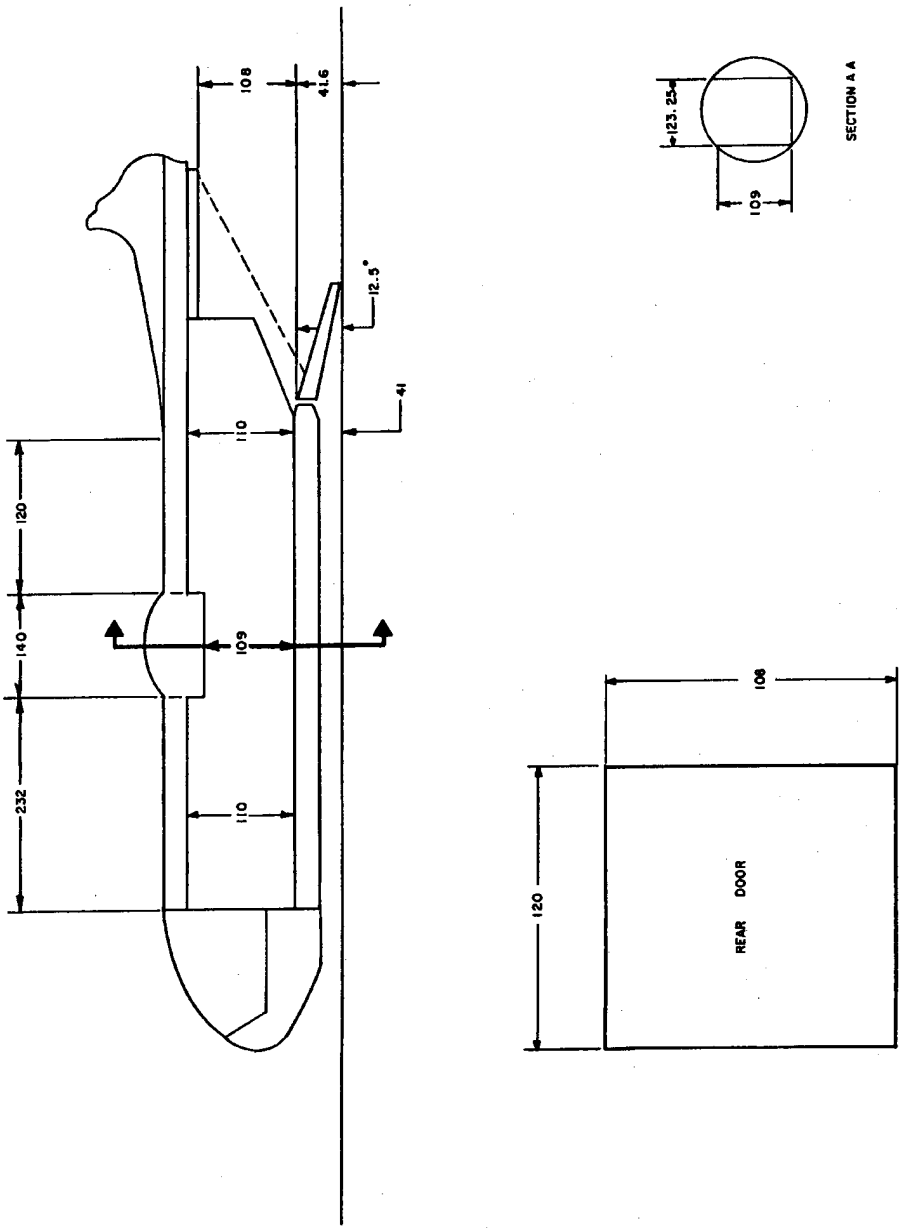


FIGURE 2. Cargo compartment profile (C-130). X-195A

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FED-STD-307	- Trucks, Light, Comm. 4 x 2, 3,000 to 11,000 lb. GVW.
FED-STD-292	- Trucks, Light, Comm. 4 x 4, 3,000 to 11,000 lb. GVW.

- Trucks having pintle height per SAE J849 for towing trailers having up to 75 percent of the truck's VIN plate GVWR; 12 volt electrical - with and without source of air for braking trailers:

KKK-T-2107	- Trucks, Med., Comm. 4 x 2, 14,000 to 21,000 lb. GVW.
KKK-T-2108	- Trucks, Med., Comm. 4 x 2, 24,000 to 32,000 lb. GVW.
KKK-T-2109	- Trucks, Heavy, Comm. 6 x 4, 34,000 to 64,000 lb. GVW.
KKK-T-2110	- Trucks, Med., Comm. 4 x 4, 14,000 to 34,000 lb. GVW.
KKK-T-2111	- Trucks, Heavy, Comm. 6 x 6, 36,000 to 60,000 lb. GVW.

- For off-road equipment prime movers, wheeled and crawler vehicles having drawbar pin height of at least 26-inches per SAE J749; no electrical or air sources for lighting or braking.

Based on the safe operation premise that any trailer should not be towed by a truck having a gross vehicle weight rating (GVWR) on the truck's VIN name plate less than 133 percent of the GVWR on the cart trailer's VIN name plate, these trailers shall be required to properly tow and perform as specified herein only behind those trucks which have a GVWR exceeding the trailer's GVWR by at least 25 percent:

Where 133 percent x trailer GVWR = minimum truck GVWR.

When and as specified by the contracting activity (see 6.2, T-6), the trailer shall be matched to the user's prime mover (truck/equipment) to insure complete, compatible, powered combination. For off-road equipment useage, the trailer chassis shall include the slow moving vehicle sign on the rear as specified in 3.9.

3.7.2 Trailer chassis mobility. The trailer chassis shall conform to the mobility characteristics specified herein when either traveling with the intended operating equipment payload, when known, or traveling with simulated payload equal to the trailer chassis' GVWR on the VIN plate. The chassis shall operate without malfunction, damage, or permanent deformation. The loaded chassis shall:

a. Travel over smooth primary paved highways at minimum 50 mph speed.
b. Travel over gravel and dirt surfaced secondary roads, haul roads, and trails up to a minimum 20 mph speed.
c. Negotiate off-road terrain, grades, side slopes, mud, sand, and snow to and from operational sites.

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d. Negotiate hard bottom creek water crossings.
e. Negotiate highway V-ditches without damage (standard FHWA Bureau of Public Roads type, 36-inches wide by 18-inches deep).
f. Withstand immobilization and tow-out operations.
g. Provide the specified minimum of suspension articulation, as cited in each specification sheet, between all combinations of wheels, front-to-rear as well as diagonal.
h. Negotiate a chatter course to produce at least 2 600 either upward or downward accelerations having a magnitude ranging between 3 to 6 g's to the center of the road wheel on the trailer chassis.
i. Trail the prime mover without bottoming, weaving, or side sway.
j. Permit turns up to a minimum angle of 60 degrees from straight ahead to the right and to the left without interference or damage of truck or trailer.
k. Withstand tractor maneuvering resulting in pivot turning of the running gear in place, both clockwise and counter-clockwise.
l. Comply with DOT FMCSR off-tracking tolerance.
m. Provide minimum braking performance in compliance with DOT FMVSS, DOT FMCSR, and the California Vehicle Code.
n. Provide braking system in accordance with TTMA RP No. 12 and RP No. 44.
o. Provide parking brakes capable of holding the applied wheels locked on longitudinal grades up to 30 percent.

3.7.3 Fording. The trailer chassis, without prior preparation shall ford hard-bottom water crossings to a depth of the loaded rolling radius of the CCE trailer's tires and 30 inches for size 9 MD trailer chassis for not less than 5 minutes duration without damage. After fording, the trailer chassis shall show no evidence of water leakage into waterproof brake components or waterproof electrical lights, and water contaimination of the wheel bearing lubricant in the hub shall not exceed a 5-percent increase.

3.7.4 Coupling and jacking. The support leg shall withstand, without damage or permanent deformation, the forces encountered in coupling and uncoupling the trailers from prime movers, and a free-fall drop of not less than 4 inches onto a concrete surface. In addition, the size 9 MD trailer chassis shall withstand manual maneuvering of the loaded trailer chassis on a hard surface by personnel. From its fully retracted position, the support leg shall raise and lower the trailer chassis with payloads throughout their full range of travel, both with and without the rear stabilizing stands, or a substitute support, in place under the rear of the trailer frame rails, without evidence of malfunctions, binding, or interference.

3.8 Human factors. The human factors characteristics of the trailer chassis shall be in accordance with following minimum requirements:

a. Comply with SAE J185 and OSHA Standards for all handholds, walkways, and stepping sufaces and include ladders per TTMA RP No. 59.
b. Provide operating and maintenance features for operation by personnel ranging from the small person, clothed, through the large person, arctic clothed, in accordance with SAE J833 and SAE J925.

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3.9 Safety. The safety characteristics of the trailer chassis, including all contractor supplied equipment shall meet the following minimum requirements:

a. Minimize hazards created by the equipment design.
b. Minimize hazards to personnel and to property.
c. Result in furnishing material having a correct safety factor.
d. Provide safety and protective devices which do not impair the operating functional requirements and furnish prominent warnings.
e. Provide correct safe guarding, enclosing, and insulating features for exposed components and systems which are subject to high temperatures, high pressures, electrical high voltage actuation or which are inherently hazardous.
f. Eliminate nonfunctional sharp edges, projecting points, and excessive length of fasteners.
g. Conform to the California Vehicle Code, Division 12.
h. Conform to DOT Federal Motor Vehicle Safety Standards (FMVSS).
i. Conform to DOT Federal Motor Carrier Safety Regulations (FMCSR).
j. Conform to the DOL OSHA Standards and the Construction Safety Standards applicable to equipment employed in construction work.

For special applications, when specified by the contracting activity (see 6.2, T-6):

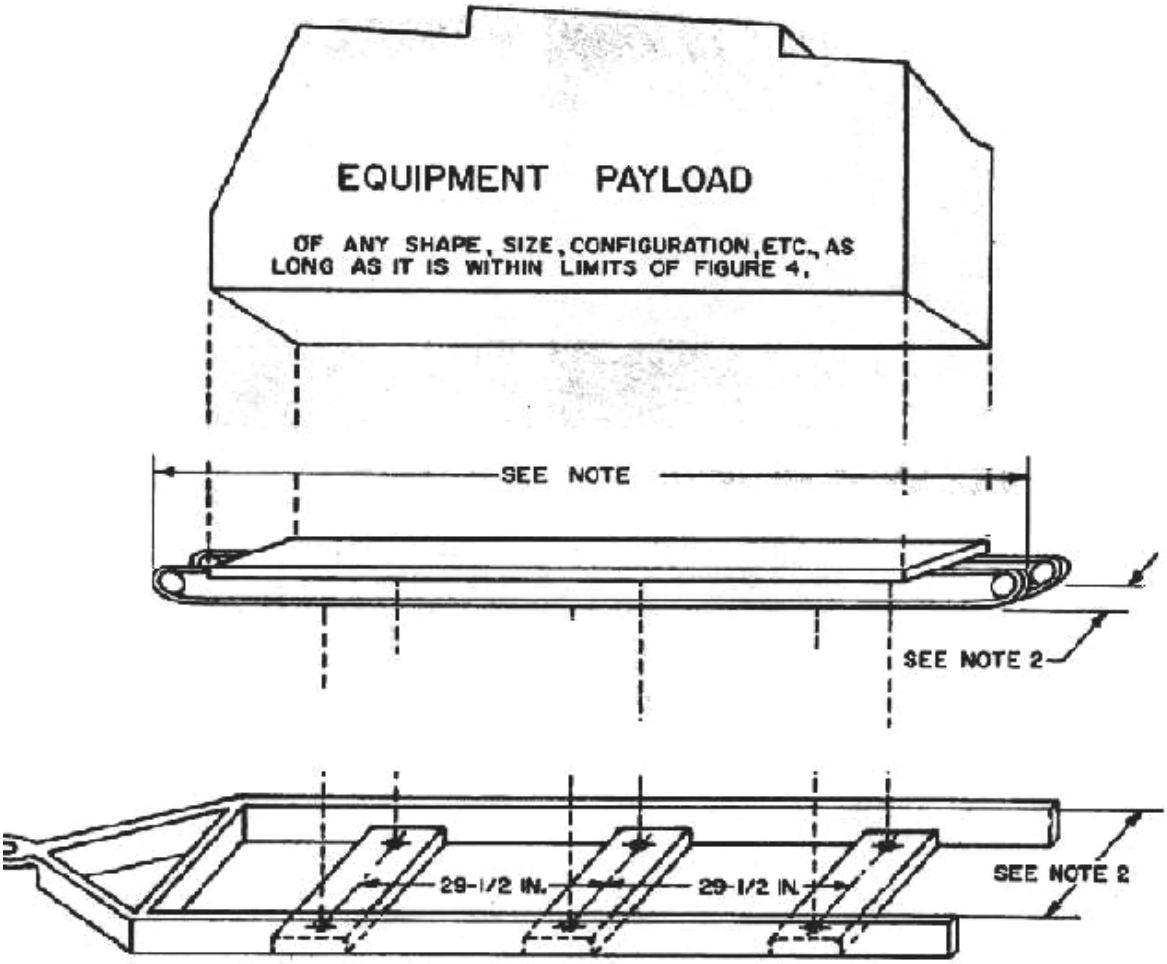
k. Provide a removable ASAE R276 slow moving vehicle (SMV) emblem on an SAE J725 mounting blade with each chassis and an SAE J725 bracket socket on the rear of each chassis for use when towed by construction equipment. Also, provide space within the toolbox for stowing and securing the SMV emblem and blade.

3.10 Environmental requirements. The trailer chassis shall be operational as specified herein in any ambient air temperature from plus 120° F to minus 40° F. The trailer chassis shall not malfunction or be damaged by water when exposed to rain ranging from a drizzle to a heavy rain of 1/4 gallon per minute per square foot of wetted surface impinging on the equipment with a 30 mph wind and when transported on a railroad car during rain conditions. During and after the precipitation, the equipment shall operate as specified herein.

3.11 Equipment mounting. The payload type, size, and configuration, for mounting various end items of supplies and operating equipment on the CCE trailer chassis, shall be as specified by the contracting activity (see 6.2, T-7) as specified herein, and on the applicable specification sheet. The intended equipment shall mount on the trailer chassis in accordance with the applicable specification sheet and figures 3, 4, 5, and 6. The equipment and chassis match shall:

a. Provide a mounting covering a minimum of 3-each chassis frame crossmembers. Provide the 3-each chassis frame crossmembers installed with the web up and flanges down, to provide a solid support for the skid mount, as shown in figure 3 and 5.

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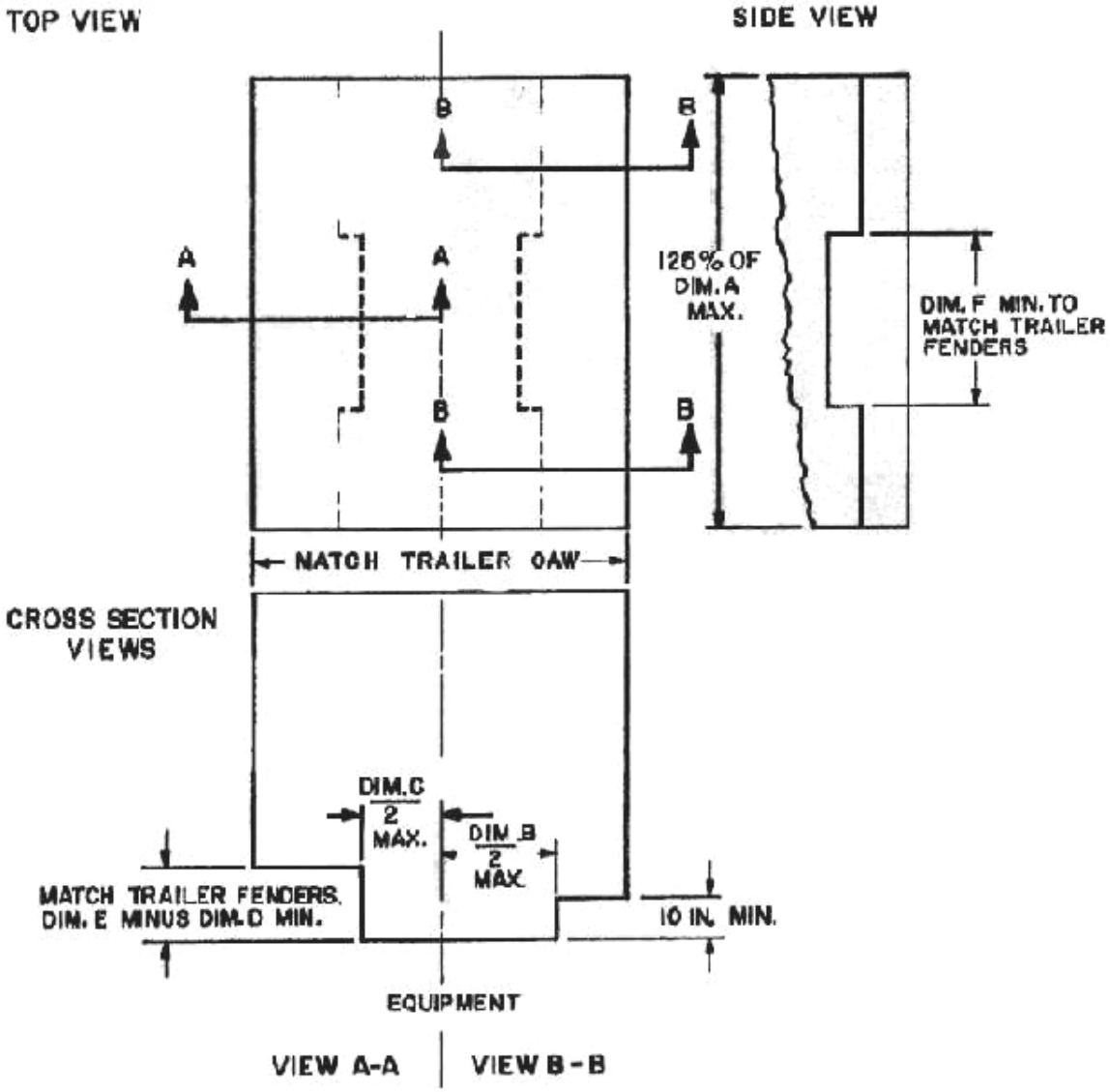


NOTES:
1. SKID DIMENSIONS PER FIGURE 5 AND SKID LENGTH OPTIONAL AS LONG AS THE SKID AND THE EQUIPMENT PAYLOAD WILL SLIP WITHIN THE CHASSIS FRAME RAILS AND MOUNT ON THE 3 STANDARD INVERTED CROSS MEMBERS.
2. FABRICATE SKID WIDTH AS NECESSARY TO BOLT DOWN TO THE STANDARDIZED CHASSIS CROSS MEMBERS WITHIN FIGURES 5 AND 6 LIMITS.

FIGURE 3. Equipment to trailer interface.

X-4057

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NOTE:
OVERALL PAYLOAD HEIGHT LIMIT 162 IN. WHEN INSTALLED ON THE TRAILER CHASSIS AND PREPARED FOR TRAVEL.

FIGURE 4. Operating equipment payload limits.

X-3072B

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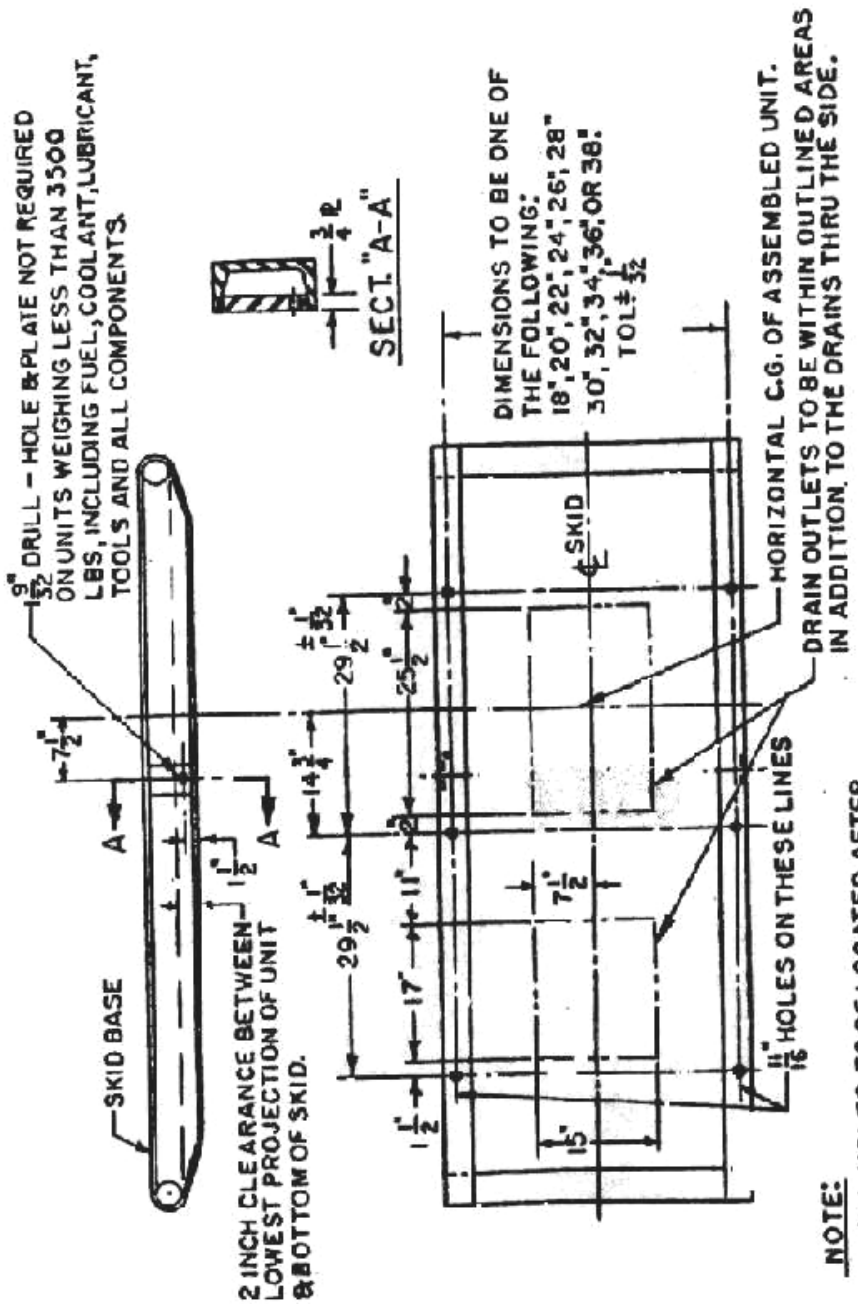


FIGURE 5. Drilling diagram & drain locations for skid-base.

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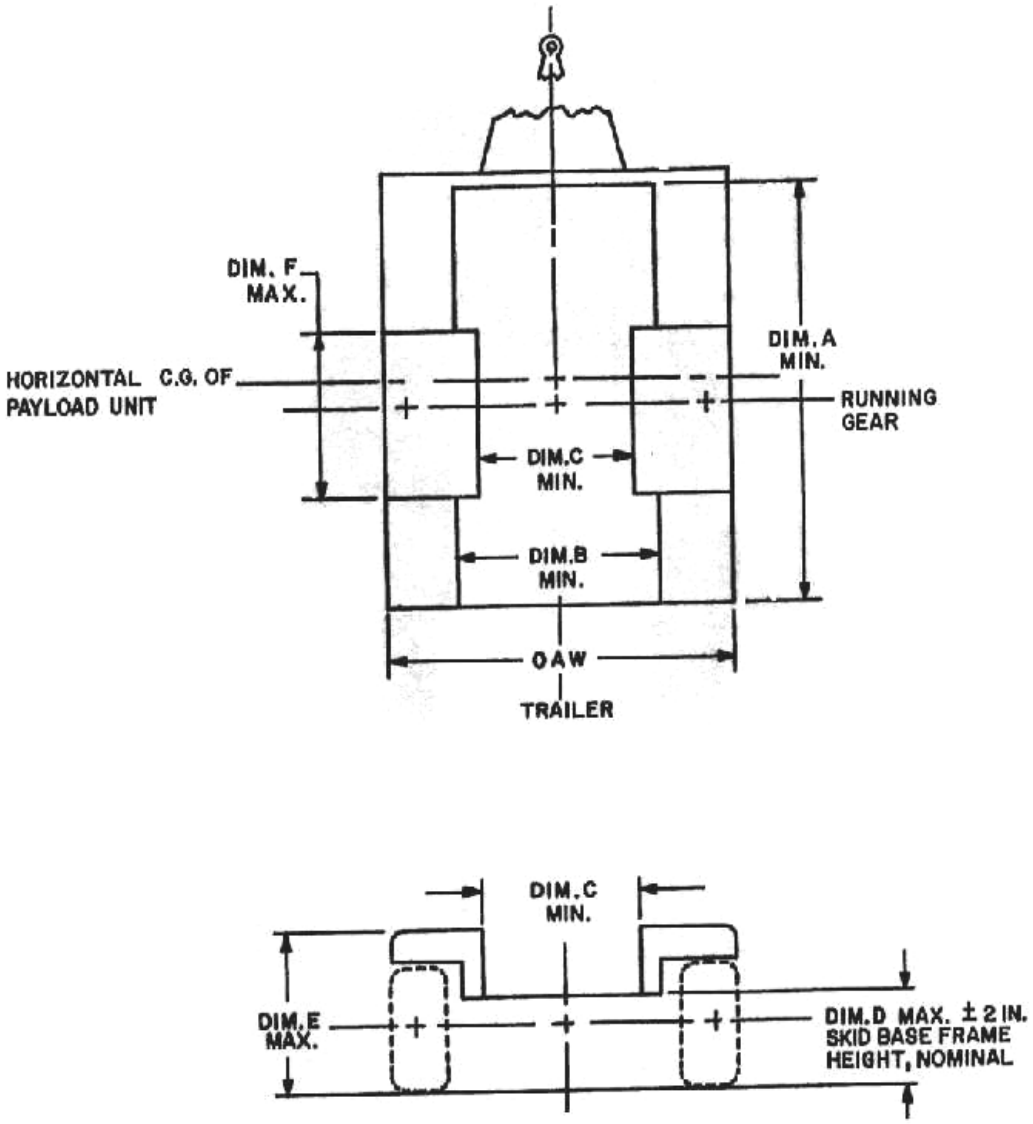


FIGURE 6. Trailer equipment bed limits.

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b. Achieve the weight on the front tow eye of the loaded trailer chassis within the allowable percent of GVWR range specified in table I of the applicable size specification sheet. Correctly locate the horizontal center of gravity (CG) of the payload, from figure 5, forward of the center of the trailer chassis running gear, as shown in figure 6.
c. By means of punch marks, of sufficient depth to remain visable after painting by the contractor, mark all equipment skid mounting hole centers on each of the 3-crossmembers to match all of the skids in figure 5. Provide punch marks symmetrical about the chassis longitudinal centerline. When specified (see 6.2, T-7), boltholes shall be drilled through all 3-crossmembers for the intended equipment.
d. Provide twin bolted-on, side shear brackets and fasteners to position and secure equipment payloads, which exceed 3500 pounds weight in accordance with figure 5.
e. Furnish drain areas for operating equipment payloads in accordance with figure 5, when open bottom skids are furnished. Where clear drain areas are not obtainable, the intrusion of chassis components shall be subject to approval by the contracting officer. When platform flooring is required with operating equipment payloads, provide space for draining coolant, oil, and other sump liquids with a common, 4-gallon-capacity drain pan having a minimum size of 27-inches long by 12-inches wide by 5-inches high, with push-pull bail type wire handle. Provide drain pan clearances for manhandling the pan per SAE J925.
f. When and as specified (see 6.2, T-7), the intended operating equipment payload and accessories shall be prepared for interface between the equipment and chassis and shall be mounted and secured to withstand all operating and traveling forces without permanent deformation or damage. When specified, provide the skid as shown in figures 3 and 5 for equipment payloads which are not skid mounted. The height of the skid shall be that dimension necessary to provide bolt installation clearances and the length shall be adequate to span all 3-crossmembers without extending forward into the towing prime mover during clearance or beyond the rear end of the trailer chassis more than 25-percent of the minimum or actual bed length, whichever is longer (total 125 percent of dimension A).
g. Provide trailer chassis and payload dimensions A through F of figures 4 and 6 as cited in table I of the applicable size specification sheet.

3.12 Trailer chassis specifications. The trailer chassis specifications shall be in accordance with the applicable size specification sheets and table II. The component sequence and number system shall be in accord with the "Vehicle Maintenance Reporting Standards" (VMRS), the trucking industry standardization system of the American Trucking Associations (ATA).

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TABLE II. Trailer specifications.

VMRS No.	Feature/Requirements
Group 10	Chassis. Provide strength, flexibility, and rigidity, to withstand dynamic transport, both highway and off-road. Withstand operational site jacking and shock and vibration imposed by the operating equipment and accessories. Provide a firm foundation for the operating equipment insuring correct alignment of machinery and distribution of weight without incurring abnormal stress concentrations. Provide components that a bolted-on in lieu of welding-on to the most practical commercial extent.
VMRS 12-001	Axle - Rear, Non-Driven. Provide single, 2-ea., or 3-ea., as specified in table I and in the applicable size of trailer specification sheet. Rate axles per DOT FMVSS and TTMA RP No. 39, and each axle manufacturer's maximum. Min. axle ratings to include the bearing and beam rating, the brake rating, and the resulting partial GAW at the hub (without tire and wheel). When and as specified (see 3.1.3, and size specification sheet, and 6.2, T-3), provide running gear for flotation as specified.
VMRS 12-001-001	Tube - Rear Axle, Non-Driven. Spindles ground to size, except where turned to size using carbide or ceramic cutting tooling with automatic machinery Steel spindles provided with hardened surfacing for bearings and seals.
VMRS 13	Brakes. Provide brake system per DOT FMVSS, DOT FMCSR, and California Vehicle Code. Provide service and parking braking means on all chassis, except none on size 1, and provide emergency-parking facilities where legally required, with the only exception that size 1 equipped only with chock blocks.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 13 (Cont'd)	Furnish brake system from tabulations A and B. When and as specified (see 6.2, T-8), furnish trailer chassis with brake systems as specified. Provide brake system components, which are exposed to road splash, with anti-corrosive treatment or protection to reduce malfunction. Where painting is applicable, clean, treat and protective-paint components per the Volume 2 Procedure of the Steel Structures Painting Council. Where inherently corrosion-resistent components are applicable, provide componenets protected with phosphate coating, min. 0.004-in. thickness of hard chrome plating, or fabricated of stainless steel.

TABULATION A. Types of brake systems.

Trailer VIN Plate Maximum GVWR	Electric (12 Volt) Brakes	Surge-Type Hydraulic Brakes	Air-Over-Hydraulic Brakes	Straight Air Brakes
Up to 2000 lb.	OPT*	STD*	OPT	NA*
Up to 5000 lb.	OPT	STD	STD	OPT
Up to 7000 lb.	OPT	OPT	STD	STD
Up to 12,000 lb.	OPT	OPT	STD	STD
Up to 15,000 lb.	OPT	OPT	STD	STD
Up to 20,000 lb.	OPT	OPT	STD	STD
Up to 30,000 lb.	VMRS 13-012, see Electro-hydraulic type only – OPT	NA	OPT	STD

*NA – Not acceptable.
STD – A standard system.
OPT – A system optionally acceptable, unless or as otherwise specified.

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TABLE II. Trailer specification. (Cont'd)

VMRS No.	Feature/Requirements		
TABULATION B. Electric foundation brake minimums, both sizes and quantities (2 per axle).			
Trailer VIN Plate max. GVWR	Min. Drum Size, 7-1/4 in, dia.	Min. Drum Size, 10-in. dia.	Min. Drum Size. 12–in. dia.
Up to 200 lb.	Min. 2 ea. size 7-1/4 x 1-1/4-in.	Opt. 2 ea. size 10 x 2-in.	Opt. 2 ea. size 12 x 2-in.
Up to 5000 lb.	Min. 6 ea. size 7-1/4 x 1-1/4-in. Min. 4 ea. size 7-1/4 x 1-1/2-in.	Min. 4 ea. size 10 x 2-in. 10 x 2-1/4-in.	Opt. 2 ea. size 12 x 2-in.
Up to 7000 lb.	Min. 6 ea. size 7-1/4 x 1-1/4-in. 7-1/4 x 1-1/2-in.	Min. 4 ea. size 10 x 2-in. 10 x 2-1/4 in.	Min. 4 ea. size 12 x 2-in. Min. 2 ea. size 12-1/4 x 3-1/2-in.
Up to 12,000 lb.	Min. 6 ea. size 7-1/4 x 1-1/2-in.	Min. 6 ea. size 10 x 2-1/4 in.	Min. 4 ea. size 12 x 2-in. 12-1/4 x 3-1/2-in.
Up to 15,000 lb.	Note: All trailer sizes below these double lines provided with adequate electrohydraulic brake systems, if and when electric brake systems are chosen by the trailer contractor or when electric brake systems are a Gov't requirement mandatory in 6.2, T-8.		Min. 6 ea. size 12 x 2-in. Min. 4 ea. size 12-1/4 x 3-1/2-in.
Up to 20,000 lb.			Min. 6 ea. size 12-1/4 x 3-1/2-in.
Up to 30,000 lb.			
VMRS 13 (Cont'd)	With the exception of wedge ramps, rollers, and S-cams, apply a thin wipe of silicone high temp compound on brake system bearing surfaces. Provides systems having ease-of-maintenance features.		

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 13-001-026	Disc - Brake. Rotors of the high heat sink type incorporating heavy-use, cooling feature design. Rotors warp-resistant when overheating occurs. Provide weatherproof puck seals which effectively seal between the puck and cast housing against malfunctions caused by road salt splash, dust, mud, water and snow and ice. Provide caliper and roter components, that are exposed to road splash, either inherently corrosion-resistance or effectively cleaned, treated, and plated.
VMRS 13-002	Rear Brakes & Drums. Provide electric, hydraulic, or straight air assemblies, as specified. Avoid components located below the axle housing and inside, between the backing plates; however, locate no components below the bottom of the brake roter or brake drum. Provide components between the drum and the backing plate or dust shields either cleaned, treated, and painted or inherently corrosion resistant. Provide dust shield protection of drum type foundation brake assemblies. With straight air S-cam drum brakes, provide the S-cam rotation and the brake drum forward rotation in the same direction on all axles.
VMRS 13-002-014	Lining - Rear Brakes. Lining selected from the FMS Automotive Data Book. Lining rated per DOT procedures and technically intended for severe, mountainous, heavy-duty commercial service. Wet lining fade not less than 65-percent of the dry friction baseline valve, per SAE J786 test code.

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TRAILER II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 13-002-014 (Cont'd)	For straight air brakes, provide lining per SAE J866, code EE, EF, or FF, having identical friction valve on all lining segments. For disc type brakes, provide linings of the semi-metallic or full-metallic types
VMRS 13-002-023	Drum - Rear Brake. Provide brake drums of the heaviest regular production optional weight available. Provide drums of the outboard mounted type to facilitate brake maintenance without hub removal, when available.
VMRS 13-002-028	Adjuster - Rear Brake Slack. Self-adjusters for each wheel providing correct compensation for lining wear without lock-up or unbalance and manual means to adjust the foundation brakes initally and in case of malfunction.
VMRS 13-004	Spring Parking Brake. With straight air brakes, provide 2 ea. (with single axle) or 4 ea. (with tandem and tridem axles) heavy duty spring type emergency-parking brake actuators on each axle, as necessary to prevent wheel rotation at GVWR on grades up to a min. 30 percent. Provide means on each trailer to cage the springs. Provide environmental protective covers, boots, and seals on each chamber.
VMRS 13-006	Brake Master Cylinder. With surge brake systems, provide master cylinder actuater and installation to comply with SAE J684. Provide surge brake actuaters per tabulation C, based on GVWR.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements			
TABULATION C. Surge brake actuators.				
Trailer VIN Plate Minimum GVWR	Minimum 6000 lb. Gross Load Actuator Unit	Minimum 12,000 lb. Gross Load Actuator Unit	Minimum 20,000 lb Gross Load Actuator Unit; and Min. 2000 lb. Tongue Load	
			Min. 1-in. Bore	Min 1-1/4 in. Bore
Up to 2000 lb.	STD*	OPT*	NA	NA
Up to 5000 lb.	STD	STD	OPT	NA
Up to 7000 lb.	NA	STD	STD	OPT
Up to 12,000 lb.	NA*	STD	STD	OPT
Up to 15,000 lb.	NA	NA	STD	STD
Up to 20,000 lb.	NA	NA	NA	STD
Over 20,100 lb.	Note: Surge brake actuator not acceptable in trailer weighing over 20,000 lb. due to likely misuse by towing with prime movers which are too light; also braking in reverse direction is required by users.			
*NA – Not acceptable. STD – A standard system. OPT – A system optionally acceptable, unless or as otherwise specified.				
VMRS 13-006 (Cont'd)	With air-over-hydraulic brake systems, furnish split hydraulic brake master cylinder, twin equallized master cylinder assemblies, or two identical power clusters. Duplicate units providing either shoe-by-shoe, brake-by-brake, or axle-by-axle service braking. Power cluster matched and approved by the foundation brake manufacturer. Provide power cluster per tabulation D.			

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements		
TABULATION D. Air-over-hydraulic power cluster assemblies.			
Trailer VIN Plate Maximum GVWR	No. of Axles	Min. Hyd. Appl. Press. @ 60 psi Air Pressure	Min. Hyd. Brake Fluid Displacement @ full Application
Up to 7000 lb. Up to 7000 lb.	1 2	700 psi 700 psi	1.8 cu. in. 1.8 cu. in.
Up to 10,000 lb. Up to 10,000 lb.	2 3	700 psi 1000 psi	2.0 cu. in. 2.0 cu. in.
Up to 15,000 lb. Up to 15,000 lb.	2 3	1000 psi 1500 psi	2.4 cu. in. 3.5 cu. in.
Up to 20,000 lb. Up to 20,000 lb.	2 3	1500 psi 1500 psi	2.4 cu. in. 3.5 cu. in.
Up to 30,000 lb.	3	1500 psi	6.0 cu. in.
VMRS 13-006 (Cont'd)		Fill all hydraulic master cylinder systems with DOT-5 brake fluid (VMRS 53-999-008) meeting the min. requirements of either FMVSS No. 116, DOT-5 or MIL-B-46176 for di-organopolysiloxane fluid (silicones).	
VMRS 13-006-008		Plug - Master Cylinder Filler. Provide Means to prevent water (during creekbed traverse) from being sucked into the master cylinder reservoir thru the filler plug breathers. Inverted U-tube breathers, having at least 6-inches of air hose on the open end, as shown in figure 7 has been acceptable.	
VMRS 13-007		Brake Lines. Hydraulic brake lines per DOT FMVSS for flex lines and SAE J1047, for steel tubing lines.	

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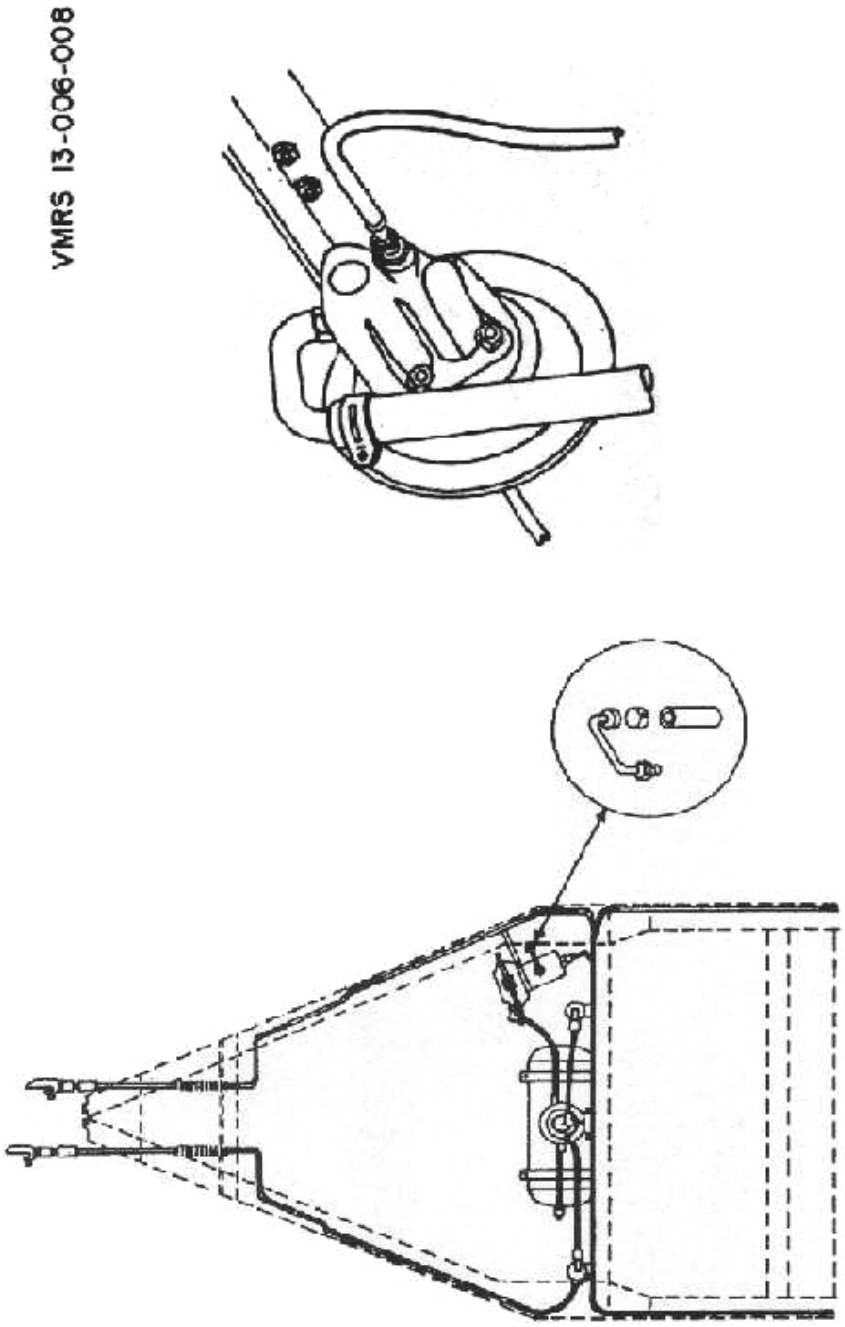


FIGURE 7. Typical U-tube breather for hydraulic brake master cylinders.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 13-007 (Cont'd)	Provide flex hose with swivel fittings furnished between master cylinder and frame and between frame member of the suspension or axle. Provide routing of the lines within the protection of the trailer framing member.
VMRS 13-010-003	Line - Air (Non-Metallic). All-hose type per TMC-ATA RP-305 or RP-307 having hose with stainless steel wire reinforcement and reuseable swivel fittings permitting hose replacement from bulk hose supply. Plumbing routed within the protection of framing members. Elastomer cushioned clamps and grommets or thru-frame connectors to secure and support hose. Support lines at a max. of 36-in. span. Min. 3-in. of hose slack at relay valves ports and identify each hoses's function adjacent to each fitting with waterproof means (tags).
VMRS 13-010-004	Tank - Air. Air reservoirs per DOT and SAE J10. Min. 2-ea. furnished, 1-ea. per axle with isolating pressure protection valves and one-way check valves and (when furnished) 1-ea. Air suspension with isolating pressure protection valve. All reservoirs furnished with interior antirust coating. No component lower than bottom of the frame rails. Employing the last, pressurized, downstream reservoir, provide plumbing and a trailer mounted hostler air valve to allow personnel to release the automatically-applied-air brakes and maneuver the trailer, without prime mover air hook-up, using only the trailer's air supply.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 13-010-004 (Cont'd)	Provide automatic reset of the hostler air valve, when the prime mover's air supply recharges the trailer's air reservoirs.
VMRS 13-010-007	Valve - Ejector, Air Tank Moisture. Provide automatic drain valves on each air tank.
VMRS 13-010-009	Valve - Relay, Air Brake. For each axle, provide 1 ea. high speed type having waterproof features, including providing an exhaust air hose long enough to preclude water entering relay valve during creekbed fording. Provide manual release, automatic reset, bleeder valve with air-over-hydraulic brake systems.
VMRS 13-010-048	Valve - Release, Spring Brake. High speed type providing selective reservoir charging, spring brake parking application, emergency application, and anti-compounding features.
VMRS 13-010-030	Air Line Coupler - Service and Emergency Gladhands per SAE J318, color coded yellow for control (service) on curbside and color coded red for supply (emergency) on roadside. Permanently connect gladhands and hose lines to trailer chassis to prevent loss. Min. hose length to reach 36 in. forward of tow eye center, compatible with SAE J849. Provide dummy gladhands to hold lines doubled back on drawbar. ID tages on drawbar connection. Locate dummy gladhands to minimize jack-knife damage to hose lines when truck and trailer are maneuvered with the lines stowed and the brakes released.

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TABLE II. Trailer specifications. (Cont'd)

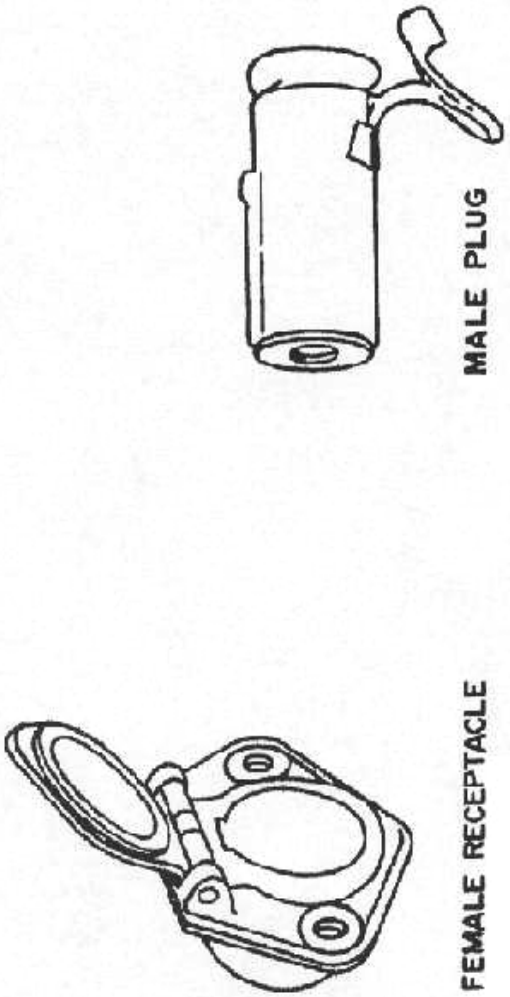
VMRS No.	Feature/Requirements
VMRS 13-010-052	Air Filter. Provide 2-ea. air filters with all straight air and air-over-hydraulic braking systems. Provide filler min. 3 x 5-in. overall size, having a replaceable filter element. In an accessible location in the drawbar area, furnish one air filter in the control line and the other in the supply line to catch and hold contaminates before they are carried over into the trailer's reservior, valves, and downstream components.
VMRS 13-012	Electric Brakes. Provide complete electrical system including driver's in-cab solid state modulated controller with hand operated wand, variable load adjustment, illuminated ammeter, wiring assembly, plug, and towing vehicle installation instructions. Electrical system for standard, 12-volt system voltage tow vehicle. Heavy duty waterproof type with vibration resistant components. Negative ground system per SAE J538. Environmental conditions per SAE J1211. Wiring per SAE J1292 and J163, with J1128 wire, and TMC-ATA RP-110 thru RP-114 components. Nut and stud or screw component terminals fastening eyelet or spade wire terminals. Wiring routed to prevent whipping, chafing, and mechanical or fatigue damage. Wiring identification per SAE J821. System waterproofed with applicable means, including encasing in flexible elastomer conduit and sealing with RTV compound and electrical varnish.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 13-012-00A VMRS 13-012-00B	Plug - Electric Brake. Receptacle - Electric Brake. Provide components and the instructions to securely install a tow vehicle electric brake receptacle on rear of tow vehicle per SAE J849. Provide trailer harness and plug assembly to reach forward min. 36-in. beyond the tow eye center. Provide clip to hold harness doubled back on tow bar. Provide electric brake receptacle and plug housings same as TTMA PR No. 40, except provide the wiring insert components of the single pole type as shown in figure 8. Provide both receptacle and plug with "Electric Brake" decal labels per 3.21.3. Locate electric brake plug and matching receptacle near the intervehicular components.
VMRS 13-012-00C	Brakes - Electro-Hydraulic. When optional electric brakes are acceptable and the electro-hydraulic system is selected, provide electro-hydraulic brakes to have the electrial portion comply with the electrical component requirments. VMRS 30, the hydraulic portion comply with the hydraulic component requirements (VMRS 13-006 & 13-007, and the brake fluid, (VMRS 53-999-008), requirements as specified for the air-over-hydraulic system. Provide 12-volt electric motor with sufficient range of operation to control and stop the trailer in a legal manner: a. Under cold region conditions, down to minus 40º F ambient temperature and a 9.0 volts at the prime mover. b. Under tropic conditons, up to plus 120° F ambient temperatures and at 14.0 volts at the prime mover.

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X-4059

FIGURE 8. Single-pole connector.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 13-012-00C (Cont'd)	Provide foundation brake assemblies selected from the air-over-hydralic foundation brake suppliers, which are matched to or compatible with the axle GAWR's.
VMRS 16	Suspension. Unless otherwise specified, provide flexibly sprung, suspension system, tailored to the load imposed and ranging between 70 to 100 percent of the axle rating, as necessary for a non-damaging ride. Provide single (1-axle), tandem (2-axle) or tridem (3-axle) as specified herein and in the applicable trailer specification sheet. Provide min. 48-l/4-in. axle-to-axle wheelbases with those multi-axle suspensions, which exceed a multi-axle actual suspension loading of 12,000 lb. and would be legally considered, by the states, as a single axle load instead of multi-axle. For off-road travel over adverse terrain provide stabilized ride of the trailer and equipment with a high center of gravity without hazardous handling and provide soft, non-damaging ride on highway. Provide suspension to protect trailer from damaging shock and vibration. Provide transverse roll angle stiffness such that a max. 7.5 degree angle, between axle centerline and trailer frame, is not exceeded at GVWR while traversing a level roadway curve producing up to 0.5 g's lateral acceleration. Provide positive control over axle rotation and tendency of the axles to brake-hop during braking. Single-axle suspension having the specified travel articulation from full compression to rebound.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 16 (Cont'd)	Provide sufficient articulation to maintain equal loading on each wheel during the min. articulation distance with max. 1 percent of GVWR allowable tolerance. Provide elastomer stops and bumpers to prevent metal-to-metal contact and damage when the axle reaches max. wheel compression position and when the axle reaches max. wheel rebound position. Suspension installation is correct alignment with tow eye and trailer centerline, per both suspension and axle manufacturers. Spring working loads rated per TTMA RP No. 32.
VMRS 16-008	Air ride. When specified (see 6.2, T-9), provide air-spring type suspension intended for on-off road use. Min. 8-in. travel range from compression to rebound each axle. On each axle, provide a transverse torsion type anti-sway bar assembly linking together the roadside and curbside trailing member and axle mounts. Provide single or multiple HD telescopic shock absorbers, at each wheel, to control compression and rebound. Provide cushioning elastomer bumper stops to prevent metal-to-metal bottoming-out of the suspension during compression. Provide positive cushioning stops other than shock absorbers to halt rebound and withstand min. 2-g unsprung weight deceleration, typical or lowering the suspended trailer with a crane. Air spring suspension of trailing member type, having piston or multiple bellows per SAE J551, internal bumpers. Provide air pressure protection valve, air controls, and air-tight plumbing.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 16-008 (Cont'd)	Ride frequency max. 130-CPM at center of the payload area. Height control valves, closed center time delay type. Control valve operating linkage protected from road splash, debris, and obstacle damage by guarding means. Time delay limits min. 3-sec. to max. 8-sec. Furnish a guarded driver's over-ride air control on roadside front within the protection of the drawbar. Air control to provide 3-positions as follows: 1. Highest maximum ride height (bags fully inflated). 2. Normal automatic ride height under control of automatic height control valve (bags mid-way). 3. Lowest minimum ride height on bumpers (bags completely deflated). As a means of reducing air consumption for trailer chassis having fixed payloads of known value, provide an additional lockout valve to trap the air pressure in the suspension air bags when closed and to provide normal, height control valve leveling when open. Install the lockout valve downstream of the height control valve. Provide plumbing of flexible hose and reuseable swivel fittings conforming to the same requirments as for the air brake system, except that the driver's over-ride air control lines may be per SAE J844, Type B nylon hose and hose fittings.
VMRS 17 & 18	Tire & Wheel Assembly. Provide the trailer tire and wheel assemblies as specified on each trailer chassis plus 1 ea. spare.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 17 & 18 (Cont'd)	Provide minimum size tires and matching wheels specified on the applicable size specification sheet and herein.
VMRS 17-001-001	Tire - New. Tires comply with TRA Year Book and DOT FMVSS for highway speeds and 50 mph for ML tires. Tires at least 100 level quality intended for use on powered units, except for 10.00R15TR trailer specials. Provide all tires with traction-road-lug type tread. Unless otherwise specified, provide 2-ea. widebase single tubeless tires on each axle, min. size or 2-ea. or 4-ea. conventional tube type tires on each axle, as specified, from the selection in tabulation E.
VMRS 17-008	Valve. Provide metal valve caps with elastomer seals for all tire assemblies.
VMRS 18-003-001	Wheel Assembly Disc. Disc wheels per DOT FMVSS, and TRA Year Book, and NWRA specifications. Tire and wheel assemblies inherently in-balance. Disc wheels heavy duty on-off road type having standard industry off-set. Wheels not to exceed max. 050-in. radial and lateral run-out (TIR) or max. 120-in. oz. out of balance, to prevent wheel tramp or drumming. Wide base single type rim and disc made by the same manufacturer or fabrication approved by rim manufacturer. Unless otherwise specified, provide 2-ea. widebase single disc wheels on each axle or 4-ea. convention dual disc wheels on each axle as specified from the selection in tabulation F.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements			
	TABULATION E. Tire Characteristics.			
Tire Size	4.80-12	5.30-12	18.5-8.5-8	LR78-16LT
Min Load Range & Ply Rating	LR C (6PR)	LR B (4PR)	LR B (4PR)	LR D (8PR)
Type of Tire	Ind, Hwy Tlr, Tube Type Bias w/ Hwy Tread	Ind, Hwy Tlr, Tube Type, Bias w/ Hwy Tread	Ind, Low Sec Ht - - - Bias w/ Hwy Tread	Conventional Tubeless Bias
No. Per Axle	2 ea., Single	2 ea., Single	2 ea., Single	2 ea., Single
Gross Tire Contact Area Per Axle	38 sq. in.	40 sq. in.	100 sq. in.	92 sq. in.
Unless otherwise specified, Conventional Use; Predominately On-Road Operation:				
Max Infl Press	90 psi	50 psi	35 psi	60 psi
Max Allowable Axle Load (Based on Tires)	1980 lb.	1590 lb.	1540 lb.	4800 lb.
When specified (see 6.2, T-3), Flotation Use; Predominately Soft-Terrain Operation:				
Max Infl Press	- - -	35 psi	30 psi	30 psi
Max Allowable Axle Load (Based on Tires)	- - -	1290 lb.	1410 lb.	3200 lb.
Design Rim Size	- - -	- - -	- - -	16 x 6.50

Note: See VMRS 57-001-001 for mandatory spare tire and mounting.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements				
TABULATION E. Tire Characteristics. (Cont'd)					
Tire Size	LT235/85 R 16	9.50 R 16.5LT	12 R 16.5LT	7.50 R 16	10.00 R 15TR
Min Load Range & Ply Rating	LRD (8PR)	LRD (8PR)	LRE (10PR)	LRD (8PR)	LRF (12PR)
Type of Tire	Metric Tubeless Radial	Wide Base Tubeless Radial	Wide Base Tubeless Radial	Conventional Tube Type Radial	Conventional Tube Type Radial
No. Per Axle	2 ea., Single	2 ea., Single	2 ea., Single	2 ea., Single 4 ea., Dual	4 ea. Dual
Gross Tire Contact Area Per Axle	____ sq. in.	102 sq. in.	140 sq. in.	S 49 sq. in. D 98 sq. in.	288 sq. in.
Unless otherwise specified, Conventional Use; Predominately On-Road Operation:					
Max Infl Press	65 psi	65 psi	65 psi	65 psi	80 psi
Max Allowable Axle Load (Based on Tires)	5200 lb.	5500 lb.	7000 lb.	S 4600 lb. D 9200 lb.	15,000 lb.
When specified (see 6.2, T-3), Flotation Use; Predominately Soft-Terrain Operation:					
Max Infl Press	35 psi	35 psi	35 psi	35 psi	60 psi
Max Allowable Axle Load (Based on Tires)	3400 lb.	4000 lb.	4700 lb.	S 2800 lb. D 5700 lb.	12,800 lb.
Design Rim Size	16 x 6.50	16.5 x 6.75	16.5 x 9.75	16 x 5.50	15 x 7.50
Note: See VMRS 57-001-001 for mandatory spare tire and mounting.					

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements			
	TABULATION F. Tire Characteristics.			
Standard Wheel:				
NWRA Wheel No.	41200 or 36316	41200 or 36316	411315	X-45203
Rim Size for Tire Size No. of Bolt Holes	12 x 4.00 4.80-12 4	12 x 4.00 5.30-12 4	8 x 7.00 18-8.50-8 4	16 x 16.50 L78-16 8
Bolt Circle	4-in.	4-in.	4-in.	6.5 in.
Max. Load Each Wheel	915 lb.	915 lb.	910 lb.	Mfgr's.
Optional Wheel:				
NWRA Wheel No.	- - -	- - -	- - -	- - -
Max. Load Each Wheel	- - -	- - -	- - -	- - -

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements				
	TABULATION F. Tire Characteristics. (Cont'd)				
Standard Wheel:					
NWRA Wheel No.	X-45203	X-45216	X-45245	X-45239	- - - 6-3/8 Offset & 7/16-in. Disc
Rim Size for Tire Size	16 x 6.50 LT235/85-16	16.5 x 6.75 9.50-16.5	16.5 x 9.75 12-16.5	16 x 5.50 7.50-16	15 x 7.50 10.00-15
No. of Bolt Holes	8	8	8	8	10
Bolt Circle	6.5 in.	6.5 in.	6.5 in.	6.5 in.	8.75 in. per SAE J694
Max. Load Each Wheel	Mfr's.	Mfr's.	2780 lb.	Mfr's.	Mfr's.
Optional Wheel:					
NWRA Wheel No.	- - -	- - -	X-45249	- - -	- - -
Max. Load Each Wheel	- - -	- - -	3400 lb	- - -	- - -

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements			
	TABULATION F. Tire Characteristics.			
Standard Wheel:				
NWRA Wheel No.	41200 or 36316	41200 or 36316	411315	X-45203
Rim Size	12 x 4.00	12 x 4.00	8 x 7.00	16 x 16.50
for Tire Size	4.80-12	5.30-12	18-8.50-8	L78-16
No. of Bolt Holes	4	4	4	8
Bolt Circle	4-in.	4-in.	4-in.	6.5 in.
Max. Load Each Wheel	915 lb.	915 lb.	910 lb.	Mfgr's.
Optional Wheel:				
NWRA Wheel No.	- - -	- - -	- - -	- - -
Max. Load Each Wheel	- - -	- - -	- - -	- - -

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 18-004	Hub, Rear-Wheel. Axle hubs for disc wheels and with mounted brake drums, when drum type brakes are furnished. Studs and stud circle per matching wheels in tabulation F.
VMRS 18-004-002	Bearing, Rear Wheel. Standardized tapered roller bearings with cups, bearing, and cones, all matched in accordance with the bearing and axle manufacturer's recommendations. Selection of the bearing sets shall be based upon the trailer chassis' min. axle rating of the appropriate specification sheet and TMC-ATA Advisory AV 6-15.
VMRS 18-004-011	Seal - Oil, Wheel. Lube oil seals per TTMA RP No. 43.
VMRS 18-004-017	Dust Cap, Rear Wheel Hub. Visual dust caps for oil lube level.
VMRS 30	Electrical Group. Provide a complete vehicle electrical system per DOT, California Vehicle Code, SAE Handbook, TMC-ATA RP Manual, and TTMA RP No. 9 and No. 14, and TTMA TB No. 69. Standard 12-volt system potential with negative ground per SAE J538. Heavy duty waterproof electrical system providing protection from shock and vibration, from physical damage, and from the environment as defined in SAE J1292.
VMRS 34-002	Lamps - Rear, Tail, Stop, Turn, License, and Back-Up. Provide total of 4 ea. combination lamps on the rear of the trailer. No license or back-up lamps required.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 34-002 (Cont'd)	Provide individual lamps to function as: - 2-ea. turn signal and clearance (double filament type). - 2-ea. stop and tail (double filament type). Combination lamps per TTMA RP No. 14 and SAE J759. Hermetically sealed lamp housing with shock mount double filaments. Round 4-in. dia. size for flush mounting in 4-1/2-in. dia. TTMA standard holes. Pigtail lamp lead to include ground wire from lamp to vehicle framing member. Install lamps as far apart as practical in rear frame members or in lamp boxes. Recess lamps forward from the rearmost edge of the trailer chassis to provide lamp protection from backing into loading docks and into trees.
VMRS 34-002-007	Reflectors - Safety. Furnish reflectors per TMC-ATA RP-702 and TTMA RP No. 9 plus 2-ea. additional on the front facing forward. Secure all reflectors to the vehicle with metal fasterners. No reflectors installed on the unprotected trailer surfaces or at an angle greater than 15 degrees from vertical.
VMRS 34-003-036	Plug (Seven Conductor Electrical Connector). Provide front intervehicular plug and cable assembly for commercial truck towing. Furnish a 7-way plug per TTMA RP No. 40 connected to a cable per SAE J1067. Permanently connect plug and cable assembly to trailer chassis to prevent loss. Plug to extend a minimum of 36-in. toward the truck from the tow bar eye center.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 34-003-036 (Cont'd)	Provide clip to hold the plug with the cable doubled back on the tow bar for stowage. Locate plug clip to minimize jack-knife damage when truck and trailer maneuver with cable stowed. Connect trailer ends of SAE wire cable into weatherproof covered, towbar mounted, electrical connection box having nut and stud type terminal block and having SAE J553 circuit brakers protecting all live trailer circuits. If furnished see 3.15, tap-in the military tow vehicle's 24-volt, 12-way plug and cable assembly wires (reduced to 12-volt) into this terminal block so that the circuit breakers will protect the 24-volt circuits the same they do the commercial 12-volt commercial circuits. Provide clip to hold the plug with the cable doubled back on the tow bar for stowage. Locate plug clip to minimize jack-knife damage when truck and trailer maneuver with cable stowed. Connect trailer ends of SAE wire cable into weatherproof covered, towbar mounted, electrical connection box having nut and stud type terminal block and having SAE J553 circuit brakers protecting all live trailer circuits. If furnished see 3.15, tap-in the military tow vehicle's 24-volt, 12-way plug and cable assembly wires (reduced to 12-volt) into this terminal block so that the circuit breakers will protect the 24-volt circuits the same they do the commercial 12-volt commercial circuits. Protectively cover, but do not enclose, heat producing devices, if furnished, which lower the voltage from 24-volts down to 12-volts.

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TABLE II. Trailer specifications. (Cont'd)

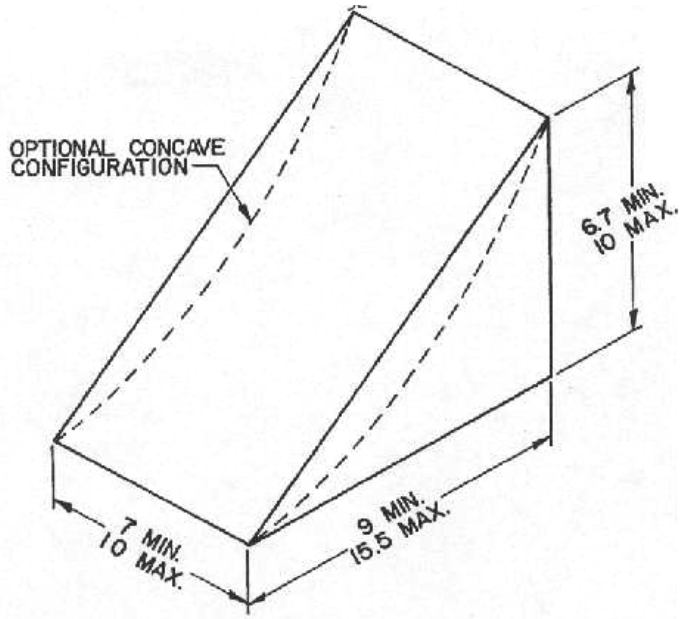
VMRS No.	Feature/Requirements
VMRS 34-004	Wiring. All wiring per SAE J163, SAE J1292, TTMA RP No. 14, and TMC-ATA RP Manual. Wire identification codes per SAE J560 and TTMA RP No. 40 or, optionally, SAE J821. Multiple circuits combined into harnesses. Wiring supported at max. 36-in. spacing with anchored elastomer ties or coated clamps to prevent whipping, chafing, and mechanical or fatigue damage. Interior of mechanical splicing devices waterproofed with RTV silicone sealant. Wire per TMC-ATA RP-110 equipped with terminals per TMC-ATA RP-112. Disconnect points for multiple circuits (in excess of 4 ea.) of the waterproof plug type or of the nut and stud juction block type. Junction blocks coated with electrical varnish after assembly. Wire-to-wire terminals secured to vehicle. For positive return ground, provide ground wires between the front receptacle, body, equipment, and chassis framing members. On a metal plate or decal, provide a vehicular wiring diagram, as actually furnished on the trailer, per TTMA Technical Bullentin No. 65.
VMRS 34-004-010	Circuit Breakers. Provide circuit brakers per SAE J553 protecting all live circuits between the trailer plug(s) and the lamps. No fuses acceptable.
VMRS 34-004-021	Lamps – Marker. All running lamps (clearance, identification, and side marker) per SAE J592 and TMC-ATA RP-108, either flush mount type per RP-108/1 or in-line protective lens type per RP-108/3.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 34-004-021 (Cont'd)	All lamps equipped with polycarbonate plastic lenses and either hermetically sealed shock resistant filaments or fleet service bulbs per TMC-ATA Advisory AV 1-2 having the Trade Number cited therein. All running lamps, clearance, identification, and side marker of same interchangeable make and model. Locations per TMC-ATA RP-702 and TTMA RP No. 9.
Group 50	Accessories.
VMRS 51-002-001	Hubodometer Assembly. When specified (see 6.2, T-10), furnish a hubodometer of the driveless, tamperproof, self-lubricating type for converting wheel revolutions into accurate mileage readings. Visual recording counter having mileage numbers in the upright postion. Min. 7-digit type recording a minimum 999,999.9-mile. Corrosion-resistant housing sealed against moisture, oil, grease, dust, and climatic conditions. Hubodometer located on curbside, axle hub. Disc wheel removal without disturbing hubodometer required.
VMRS 53-004-013	Chock – Wheel. Provide 2 ea. chock blocks and cradles for single axles and 4 ea. chock blocks and cradles for multi axles, located 1 or 2 on each side near the forward rear, axle(s). Provide chock block per figures 9 and 10. Captive chains of a length to place blocks both ahead of and behind the forward tires of the front axle without being taunt. Cradles installed on a vertical flat surface with ears up and the lower loop down having excess chain figure-eight-wound over the ears.

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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEIGHT- 11.5 LB.

CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 9. Chock block assembly.

X 603 A

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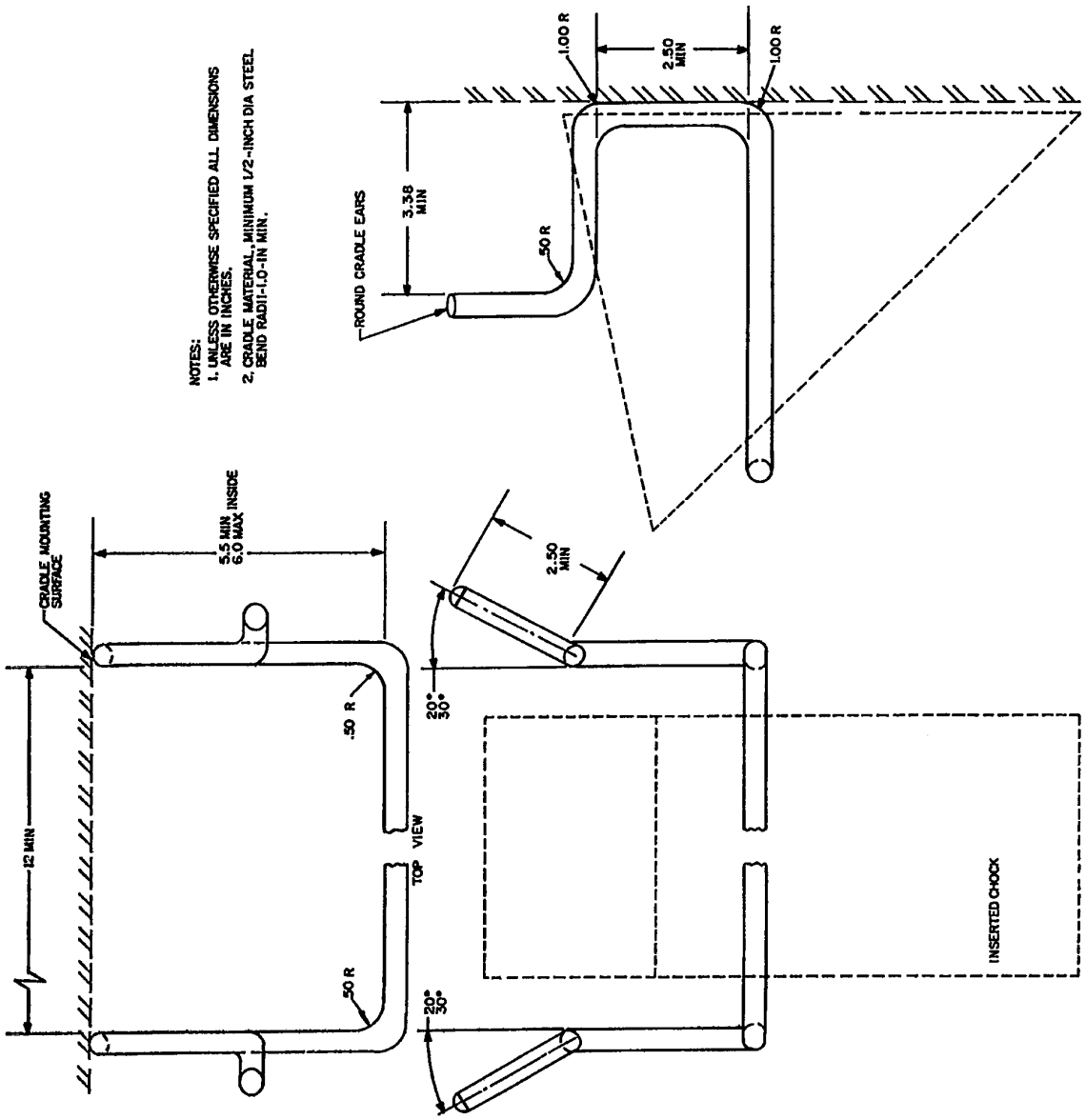


FIGURE 10. Cradle.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 53-999-008 & 53-999-009	Fluid - Hydraulic Brakes. Fluid - Hydraulic Clutch. Service the vehicle's hydraulically actuated components, which normally use non-petroleum based, glycol brake fluid as a medium, with DOT FMVSS No. 116, DOT 5 type fluid meeting the minimum requirements of MIL-B-46176 for di-organopoly-siloxane fluid (silicones). Includes all non-petroleum based fluids for controls, as well as air-over-hydraulic disc brakes.
VMRS 57	Spare Wheel Mounting.
VMRS 57-001-001	Carrier - Spare Wheel. Provide a spare wheel mount, complete with an inflated spare tire and wheel assembly of the same size as specified for the road wheels (VMRS 17-001-001, etc.).
VMRS 57-001-004	Lock - Spare Wheel. Provide locking means acceptable for a minimum 3/8-in. padlock to prevent loss of the spare.
VMRS 57-001-005	Bracket - Spare Tire Mounting. Install the spare tire and wheel assembly bracket secured either to the No. 1 trailer crossmember or to the side rails of the frame so as to avoid interference with jack-knife turning of the towing vehicle and yet not intrude into the allotted payload area for the operating equipment (see 3.11).
VMRS 59	Vehicle Coupling System.
VMRS 59-003	Tow Hooks. Provide 2-ea. tow hook and safety chain assemblies per DOT FMCSR [49 CFR 393.70(f)] and SAE J697. Provide safety type chain hook on truck-ends of chains.

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TABLE II. Trailer specification. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 59-003 (Cont'd)	Install so safety chains extend a min. 30-in. forward of the center of the tow eye, toward the towing truck.
VMRS 59-004	Drawbar. Provide A-frame or pole type trailer drawbar on front of trailer. Min. drawbar length, measured from center of tow eye to No. 1 trailer crossmember, per table I of the applicable size specification sheet. Provide A-frame angle to permit towing truck of 96-in. width to turn min. 60-degree right and min. 60-degree left, with respect to the trailer, without interference, except for size 1 and size 9. Provide size 1 compatible with 74 in. wide towing truck. Provide size 9 with min. 40 degree turning angles when towed by GI trucks.
VMRS 59-007	Lunette Assembly. Provide tow eye per SAE J847. Provide tow eye of alloy steel having a case hardened wear surface for at least 120 degrees on the forward, top and bottom of the eye. Locate the center of the tow eye within a max. tolerance of 1/4-in. from the longitudinal centerline of the trailer. Except for sizes 1 and 3, provide multiposition vertical range of tow eye adjustment from at least 26-in. height per SAE J749, for crawler tractor drawbars, thru at least 33-in. height per SAE J849, for truck and trailer pintle hooks. Provide range of 20 to 25 in. lunette height for size 1 and 3 chassis. Provide change of height adjustment, of chassis 4 and larger, by a mechanic using only common mechanic's tools per 3.7.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 72-001-004	Bumper Assembly - Complete. Provide a rear bumper assembly protecting the full width of the rear of the trailer body, equipment, and chassis. Provide min. bumper height of min. 8-in. Provide underneath metal skid plate which functions to prevent the bumper from hanging on rock and stumps or bulldozing dirt while crossing highway V-ditches.
VMRS 77	Trailer Frame (Including Support)
VMRS 77-003	Leg Assembly. Front support leg of heavy-duty single leg type having hand crank and holder, bevel gear box, anti-friction travel screw, telescopic leg cover fully enclosed, and self-aligning sand shoe. Provide support leg of either telescopic multistage type with min. 20 in. retraction travel range or folding type trunnion mount type having min. 14-in. travel range and then swing-up. Provide clearance for V-ditches without hang-up. For size 1 and 3, provide min. 12-in. vertical leg travel. For larger sizes, provide min. 20-in. vertical leg travel or travel and swing-up. Min. rated front leg capacity 60 percent of total trailer's GWR on the VIN plate. All-weather-type, silicone base lubricant in leg mechanism. For size 1 and 2, provide min. 9-in. road clearance when ready for travel. For larger sizes, provide min. 12-in. road clearance when ready for travel per SAE J849 max. pintle height.
VMRS 77-003-004	Sand Shoe. Provide equalizing sand shoe having turned-up edges. For size 1 and 3, provde min. 32-sq. in. projected area for each shoe. For larger sizes, provide min. 64-sq. in projected area for each shoe.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 77-012-002	Frame Assembly – Complete. Frame assembly per DOT FMCSR 393.71(h) with drawbar and tow eye. Frame of all-welded construction with main longitudinal frame rail members of the continuous type from front tow eye to rearmost crossmember. Frame material of at least ASTM A242 or SAE J410 HSLA steel having min. 50,000-psi guaranteed yield point (yp). No permanent deformation of frame with 175 percent of rated payload (GVWR minus curb weight) applied to the 3-each equipment mount crossmembers in 3.11 or, for size 32 trailer, inmost critical 8-linear-feet of the platform, when supported at tow eye and suspension mounts on frame and when supported at the tow eye and the rear jack stands (for operating equipment leveling). When specified (see 6.2, T-11), for cold region operation, fabricate frame assembly of at least ASTM A710, min. Grade A, having nickel and copper additives suitable for carrying payloads dynamically in low arctic ambient temperatures.
VMRS 78 VMRS 78-002 & 78-009	Trim & Miscellaneous Hardware. Fender - Rear. Deflector Assembly - Gravel. Provide rigid rear fenders and flexible splash guards for trailer rear running gear. Wheel splash protection per SAE J682 and J683 and California Vehicle Code. Lightweight FRP or aluminum rear singles tandem, or tridem fenders to withstand min. 300-lb. load on any l-sq.-ft. area at any one time. Splash guards per TTMA RP No. 41 and TMC-ATA RP-701, of anti-sail type.

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 78-009	Flexible reinforced elastomer having min. 3000 psi ultimate strength and inherent cold-weather resistance. Provide splash guards having an integral textured splash surface of short blades or pins. Mount TTMA splash guards to brackets and to fenders by metal clamping strips fastened with hex bolts and nuts thru the TTMA/TMC-ATA standard hole pattern. Angle each guard 5- to 10-degrees inward toward the center to reduce water mist during rainy weather.
VMRS 79	Safety Devices.
VMRS 79-00X	Rear Supports. To prevent tipping backwards, when the trailer chassis is parked for equipment operation, furnish a pair of commercial, trailer stabilizing jacks. Permanently fasten trailer stabilizing jacks to frame. Provide jacks of the inverted scissors type having a swing-down, screw-actuated push arms and ground contact pads. Offset handle hex crank for jacks furnished with each trailer. Furnish jacks of the type that self-store for travel by retracting up to the frame. Individually adjustable for side slopes up to 5 percent with trailer level. Each jack or the sum of each two jack rating on each side equal to a total jack rating min. 40 percent of trailer's GVWR on the VIN plate. Drive each two jacks, on a side, together. Withstand static load and dynamic equipment operation with trailer supported only by front support and by both rear stabilizing jacks. When self-stored for travel, locate jacks so as

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TABLE II. Trailer specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 79-00X (Cont'd)	to not decrease the trailers angle of departure at the rear and not hang-up on V-ditches. For size 9 chassis, furnish the twin garage axle stands shown on the drawings and store them in the front. For size 32, provide 2 ea. removeable screw jacks per GGG-J-51, Type 1, Class 3. Provide jacks having min. 2-1/2 in. dia. screws, steel ball bearing caps, auxiliary base plates to reduce loading below 80 psi. Provide chassis levels to permit personnel to jack-up and level the operating equipment for level operation on terrain having a slope of up to 5-percent in any direction. Provide min. vial bubble level recessed on outside corners of van body. 1-ea. front and 1-ea. rear for leveling longitudinal axis. l-ea. front and l-ea. rear for leveling transverse axis. Visibility of levels to permit personnel operating the leveling jacks to accurately level the body for operation. Levels waterproof as installed and recess mounted to prevent damage.

3.13 Trailer body specifications. When and as specified (see 6.2 Appendix, T-12), a trailer body, in part or whole, shall be furnished. The body shall be skid mounted on the trailer chassis as specified in 3.11, except for size 32 where the platform body is intergral with the trailer frame. The CCE body components, as specified, for the CCE sizes of trailer chassis shall be in accordance with table III. For size 9 MD trailer chassis, the bodies shall be as covered on the drawings in the Water Purification, Distillation, or Lube and Servicing Unit Specifications (see 6.1). The component sequence and numbering system shall be in accordance with TMC-ATA VMRS.

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TABLE III. Trailer body specifications.

VMRS No.	Feature/Requirements
VMRS 34-001-0XX	Lamps - Working, Operational. As specified, provide min. 2-ea. sealed beam work lamps on the trailer body to illuminate the work area. As shown in figure 11, provide work lamps per SAE J598, 12-volt sealed beam unit, PAR 46, Trade No. 4478, installed in shock mount rubber ring, construction-equipment-type lamp housing. Mount lamp housing on the trailer body. Intall 1 ea. lamp on curbside and 1 ea. lamp on roadside, as far apart as possible and as high as practicable without any part of these lamps being outside of the trailer body perimeter. Mount lamps in accordance with SAE J589. Aim the two inboard or lower lamps to illuminate the rearward working area. Provide SAE J533 circuit breakers, min. 14GA SAE J1128 Type STS or HTS wire, and SAE J561 spade terminals throughout. Provide labled floodlamp switches on trailer body. Min. 20-amp lamp rating capacity of each switch. Provide driver control or the work area by a switch. Provide adjacent amber (yellow) indicator lamps for each switch, warning that the lamps are “ON”.
VMRS 34-004	Wiring. Provide wiring to feed the work lamps from the engine alternator battery d.c. electrical system of the operating equipment, if available, or otherwise from the Auxilliary Power Circuit of TTMA RP No. 40 intervehicular plug, circuit BLU, No. 7. All wiring per SAE J163, SAE J1211, SAE J1292, TTMA RP No.14, and TMC-AT ARP Manual. Wire identification codes per SAE J560 or, optionally, SAE J821. Multiple circuits combined into harnesses.

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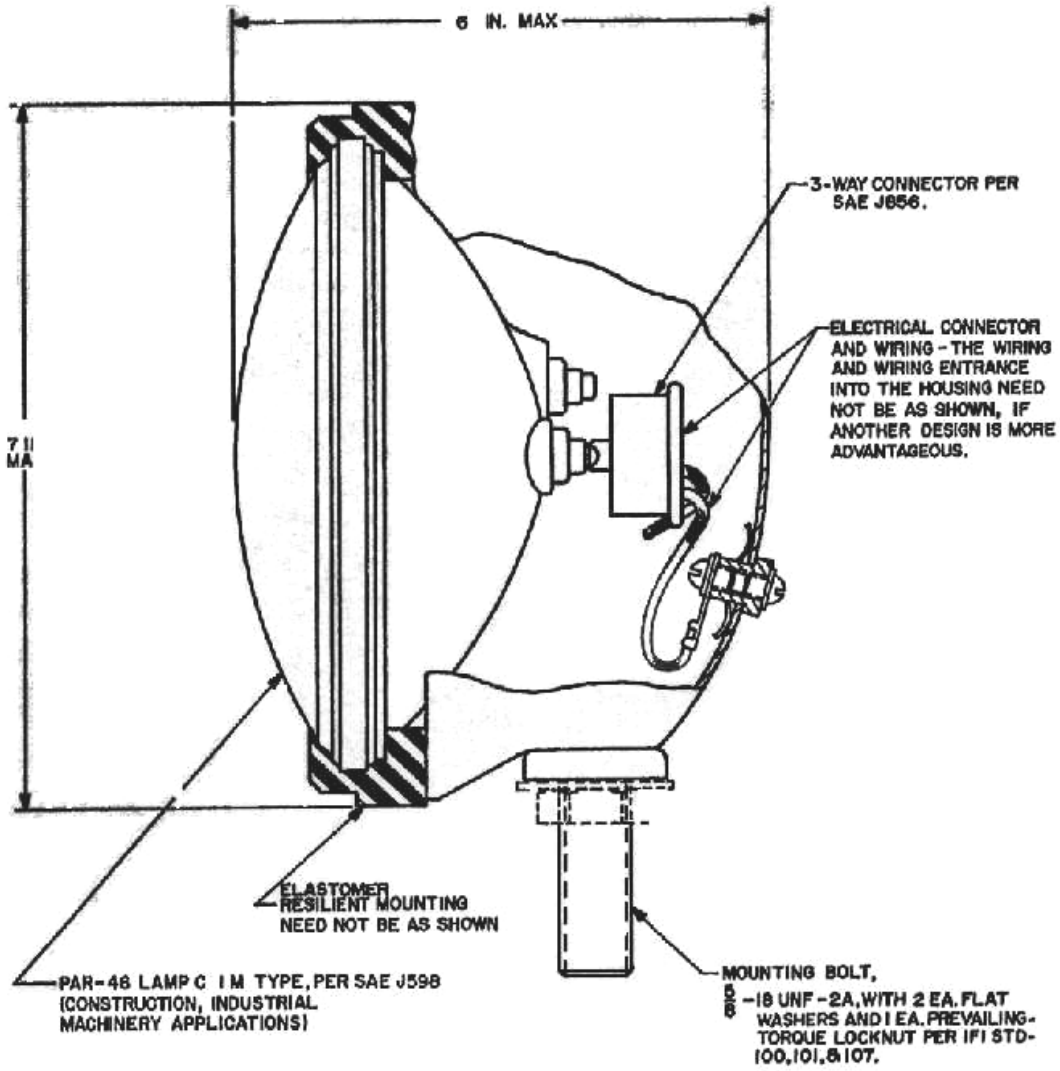


FIGURE 11. Housing, lamp, resilient mounting.

X-250A

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TABLE III. Trailer body specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 34-004 (Cont'd)	Wiring supported at max. 36-in. spacing with anchored elastomer ties or coated clamps to prevent whipping, chafing, and mechanical or fatigue damage. Interior of mechanical splicing devices waterproofed with RTV silicone sealant. Wire per TMC-ATA RP-110 equipped with terminals per TMC-ATA RP 112. Disconnect points for multiple circuits (in excess of 4 ea.) of the waterproof plug type or of the nut and stud junction block type. Junction blocks coated with electrical varnish after assembly. Wire-to-wire terminals secured to vehicle. For positive return ground, provide ground wires between engine, body, and any other component not assured of a reliable metal-to-metal electrical pathway.
VMRS 34-004-010	Circuit Breakers. Provide, circuit breakers per SAE J553 protecting all circuits. Fuses and fusible links not acceptable.
VMRS 34-004-020	Housing - Dome Lamp. As specified, inside the body provide at least 2-ea. dome lamps with accessible switches, one in each end of the body. Locate the switch inside, near the personnel door. Dome lamps same as rear vehicular combination lamps except for single filament and clear or frosted lens. Dome lamps flush mounted in standard 4-1/2-in. holes per TTMA RP No. 14. Wire the dome lamps into the front trailer circuit breaker for the Auxiliary Power Circuit, BLU, No. 7, of the TTMA RP No. 40 intervehicular plug.

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TABLE III. Trailer body specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 71	Body
VMRS 71-001	Floor Assembly. As specified, provide a full width wood and aluminum or steel platform body for the equipment specified by the contracting activity. Provide full width crossmembers integrally welded into longitudinal frame rails and the platform perimeter frame. Cover platform with a replaceable, weather resistant treated wood plank flooring. Plank min. 2-in. nominal lumber thickness. Pressure-test all planks to comply with the American Wood Preservation Association's (AWPA) Standards for resistance to rot and decay. Wood planks 1/4 to 1/2-in. above steel framing. Secure planks with plated recessed bolts and nuts, relying on bridge clips to clamp the flooring to the undrilled frame longitudinal and crossmembers. As specified, substitute min. 1/4-in. aluminum tread plate floor covering for the wood planking.
VMRS 71-002 & 71-003	Front Wall or Headboard Assembly. Side Wall Assembly. As specified, cover all stationary walls of the body with exterior paneling of the smoother skin type, waterproof, airtight, and stress-wrinkle free. Panel material of either fiberglass reinforced plastic (FRP) with plywood core or aluminum sheeting intended for containers and trailers. Provide FRP plywood per APA Plywood Design Specification Manual X215 and U.S. Product Standard PSI. Core of APA Group I, Exterior Type. Panels min. 1/2-in. thick on front,

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TABLE III. Trailer body specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 71-003 (Cont'd)	sides and roof panel having laminate of min. 22-oz./sq. yd. material or woven roving (W.R.) or combination of both. Ultra-violet-inhibitor gel coat on exterior of panel having min. 0.015-in. thickness. Exposed edges of plywood treated or dipped to repel moisture. Aluminum sheeting min. 0.05-in. thick, smooth or horizontal rib trailer type. Inside surface of the skin correctly degreased and coated with primer or adhesive to insure foam insulation adhersion.
VMRS 71-003-028	Logistic Tracks. As specified, provide commercial interior logistic tracks, vertically mounted on the body side walls. Industry standardized "Type E" provided, unless "Type F" are required. Each track spaced on 16-in. on-centers from the rear to the front and from floor to ceiling of side walls. Delete tracks at window and door openings. Each track secured to provide a mount for operating equipment weight load of min. 200-lb. having a center of force at 12-in. out from the walls and 24-in. below the ceilings.
VMRS 71-004	Roof Assembly. Roof material same as wall materials. Seamless, l-piece roof sheet having adheseive bow fastening. Provide roof crown for water drainage. Fasten roof around the perimeter only and seal fastener penetration to prevent moisture entering van body. Withstand exterior roof load, min. 660 lb./sq. ft. over any 2-sq. ft. area at anyone time and min. 40 lb./sq. ft. snow load over the whole roof.

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TABLE III. Trailer body specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 71-005	Side Door Assembly – Hinged. As specified, provide hinged door and panel assemblies as required for the operating equipment panel assemblies; swing up awning panels; accordian panels, and other hinged panel features. Door hardware of noncorrosive marine container type. Air seal flexible gaskets of silicone elastomer type with abrasion protection and installation not subject to damage by personnel foot traffic. Personnel doors equipped with hold-open catches and door bumpers. Anti-drip rain protection over each personnel door. Provide personnel doors having a size min. 30-in. wide by min. height as specified or 69-in. high. Door centered approximately 80 percent of the interior length from rear of body. Door hung to open outward and hinged on forward exterior edge. Hinge of full height piano type having min. 1/4-in. stainless steel hinge pin. Roller-on-ramp latching mechanism featuring upper and lower securing. Inside and recessed or flushed outside door handles. Door lockable from outside by a padlock having a min. 3/8-in. dia. shackle.
VMRS 71-008 & 71-009	Lining - Flat Fiber Glass. Lining - Moulded Fiber Glass. As specified, apply interior lining panels of the pebble or of the smooth flat refrigerator FRP type. Panels having laminate of min. 18 oz. per sq. yd. material or woven roving (W.R.) or a combination of both. Ceiling, front, and sides min. 0.187-in. thick FRP sheet. FRP panels of the low-fire-hazard type having a min. UL class B fire hazard classification. Panel features per ASTM E-84.

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TABLE III. Trailer body specifications. (Cont'd)

VMRS No.	Feature/Requirements
VMRS 71-009 (Cont'd)	- Max. 20 flame spread. - Max. 0 fuel development. - Max. 200 smoke development. - Less toxic than Red Oak. All panel joints and edges sealed to prevent entry of moisture into the insulation using silicone RTV sealant.
VMRS 71-019	Insulation. As specified, insulate body with closed cell, rigid form, such as isocyanurate, polybenzimidizole, or polyurethane foam. Fill and bond space, without voids, between exterior and interior panels. Nominal thicknesses min. 4-in. roof, min. 3-in. sides and front, and min. 2.7-in. under floor. Insulation shall be rated class A material and shall have a maximum flame spread index of 25 in accordance with ASTM E-84. Foam to adhere to body surfaces with an adhesive force greater than the cohesive strength of the insulation. After foam-in-place, max. 5/8-in. of bow and warp of exterior and interior panels when measured with a min. 4-ft. long metal straight edge.

3.14 Chassis components. When and as specified by the contracting activity (see 6.2, T-13), furnish chassis components with the trailer as required to assist equipment operation, including:

- Tool boxes.
- Folding catwalks and handrails.
- Weather enclosures.
- Power sources, water, fuel, etc.
- Shelving.
- Blackout lighting kit per Army Guide

3.15 Military vehicle electrical system. When specified (see 6.2, T-14), the trailer shall be additionally equipped with a military compatible, 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the

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TTMA 7-way plug, the trailer shall be equipped with the following 12-way military intervehicle components:

a. Military 12-Contact Plug – MS75020-1.
b. Military Intervehicular Cable – MIL-C-13486 and MIL-C-13486/12.

The military plug and cable assembly shall be the same length as the TTMA plug and cable assembly and both shall be labeled with identification plates for 24-volts and 12 volts respectively, negative ground. The military 24-volt intervehicular cable shall be routed into a weatherproof stud and nut type junction box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. As required for the TTMA plug and cable, see VMRS 34-003-036, the military plug and cable assembly wire ends shall terminate in spade type connectors of TMC-ATA RP-112. The assembly shall be permanetly fastened to the trailer by strain relief clamp and the terminal connectors to prevent detachment and loss of the plug and cable. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable solid state devices or resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with table IV. The resistors shall be exposed to ambient air for cooling and shall be protected from damage, road debris, and accidental burning of operational personnel.

TABLE IV. Military-Commercial Connections.

Function	Mil. Plug Terminal	Connections Instructions	Comm. TTMA Wiring Ident.
Blackout Marker	A(24)	No.	–
Ser. Left Turn & Stop	B(22-461)	Connect to –	YEL (3)
Blackout Marker	C(24)	No.	–
Ground	D(90)	Connect to –	WHT (1)
Ser. Tail & Marker	E(21-489)	Connect to –	BLK (2)
		& Connect to –	BRN (6)
Blackout Stop	F(23)	No.	–
Blackout Marker	H(490)	No.	–
Ser. Right Turn & Stop	J(22-460)	Connect to –	GRN (5)
Aux. Power (Hot)	K(37)	No.	BLU (7)
Ground	L(90)	Ground to Frame	–
Electric Brake	M(53)	No.	–
Electric Brake	N(53)	No.	–

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically insolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger

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car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the trailer's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the trailer's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.16 Fungus and moisture resistance. The electrical circuitry, including all componets and connections, shall be protected form the effects of moisture and fungus growth by an overall treatment with varnish conforming to MIL-V-173 or a varnish having 1 percent of copper-8-quinolinolate (by weight) based on the nonvolatile content of the varnish, with the following exceptions.

a. Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component and element sufaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.17 Electromagnetic interference. The electromagnetic interference emission characteristics of the equipment shall conform to the recommended limits specified in SAE J551.

3.18 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating as specified herein. Lubrication intervials shall be in accordance with SAE J752, as recommended by the manufacturers. The equipment shall operate as specified herein when lubricated with lubricants selected from SAE J754 and, when practical, shall be interchanged or be replaced by military lubricants (see 6.8). All initial fills of lubricants shall be limited to the types approved by the component manufacturers. Lubrication fittings shall conform to SAE J534 and shall be of the threaded type. Fittings shall be located in a protected position and shall be accessible to a grease gun. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. An automatic pressure-release fitting shall be provided where the use of pressure lubricating equipment could damage grease seal or other parts. Enclosures which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with fluid level indication means. A metallic lubrication chart, conforming to SAE J753, shall be furnished on the drawbar or number 1 crossmember. The chart shall conform to 3.21.3.

3.19 Servicing and adjustment. The contractor shall service and adjust the equipment for delivery in operational condition for the user. The contractor shall use his commercial checklist and include at least the following:

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- Adjustment of machinery to travel positon, including stowage and securing all plant components.
- Adjustment of suspension.
- Adjustment of brake system.
- Electrical system continuity checks and voltage drop check.
- Torque of wheel lug nuts.
- Inflation of all tires.
- Lubrication checks and lubrication of all trailer components with approved products for severe service.

3.20 Finish treatment and painting. The equipment including all compoents and parts shall be cleaned, treated, and painted in accordance with MIL-T-704, Type F, or, when specified (see 6.2), the Steel Structures Painting Council (SSPC) Volume 2, "Systems and Specifications", Specification SSPC-5, SSPC-6, or SSPC-10. The final color shall be lusterless forest green in accordance with MIL-C-46168, Type II. Inaccessible areas of the equipement shall be treated with antirust coating material in accordance with TTMA Technical Bulletin No. 56. After painting, the portions of the equipment fenders exposed to road wheel splash shall be undercoated using a commercial, sandless, undercoating material. Wheels, axles, suspension, and moving components shall be kept free of undercoating.

3.21 Identification data plates. Data plates shall be applied to the equipment on metal plates in accordance with the "Metal Name Plate Industry Standards and Practices", as published by the National Association of Metal Name Plate Manufacturers, Inc., (NAME). Identification and shipping data plates shall be of material in accordance with MIL-P-514, Composition C. The plates shall be secured to each end item in accordance with the NAME Standards and Practices.

3.21.1 Identification. The equipment shall be identified in accordance with DOT, the manufacturers, and MIL-STD-130 on MIL-P-514, Type I, Style 1 plates. The date of delivery shall be that of the complete equipment under contract and shall not be prior to date of Government acceptance.

3.21.2 Shipping data. The equipment shall have shipping data applied to MIL-P-514, Type III plates. These plates shall show the silhouette of the end items, in transport positon, indicating the center of gravity and the equipment manufacturer's approved points capable of being lifted or tied-down or, when furnished, the location and capacity of the lifting and the tiedown attachments. The plates shall be attached in a conspicuous protected location on the curbside.

3.21.3 Instruction. The equipment shall be equipped with instruction plates, decals, or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing. The safety, warning, caution, and notice plates shall be in accordance with ANSI Z35.1 and compatible with the space available.

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3.21.4 Wiring diagram. The electrical wiring diagram plates or decals shall be furnished on the equipemnt, following TTMA Technical Bulletin No. 65 for the trailer and SAE J821 for the power unit.

3.21.5 Operating instruction. Operating instruction plates or decals shall be furnished and shall be permanently affixed in a readable location near the applicable components. Instructions shall be clear and concise and shall contain sufficient information to enable operation without damage to the components or injury to personnel.

3.21.6 Trailer safety brake marking. When straight air brakes are provided for the trailer chassis, a decal inscribed with the legend reading similar to "CAUTION – Towing vehicle air lines must be connected and air control valve opened or each chamber spring annually caged before trailer parking brakes will release. Except during an emergency, DO NOT transport or park trailers with the spring brakes caged" shall be affixed on the front of the trailer in the vicinity of the gladhands. Legend shall incorporate minimum 1/8-inch-high lettering. Lettering and color shall be as specified in 3.21.3 for ANSI requirments. In addition, a caution decal describing the practice of caging of the spring brake chamber during prolonged plant operation, shall be furnished in the same general area.

3.21.7 Engine sound level requirement. When the equipment includes a diesel engine, the engine shall be furnished with noise reduction means including acoustical package and other optional equipment, when available. The noise reduction means shall reduce the mean work cycle sound level with operator present, to not greater than 85 dbA, as measured in accordance with the SAE Recommended Practice, work cycle procedure, or J919. If the sound level exceeds 85 dbA, a warning plate as specified in 3.21.3 shall be affixed to the instrument panel and shall be worded as follows:

"WARNING
EAR PROTECTION REQUIRED
WHEN THE EQUIPMENT IS OPERATING"

The plate coloring shall be the yellow and black, ANSI requirements.

3.22 Vehicle marking. As specified by the contracting activity for the appropriate service (see 6.2 and 6.7), the registration numbers and other markings of the equipment shall be in accordance with MIL-STD-642. The GVWR shall be marked on each forward side of the equipment in manner to be discernible. The TRA Year Book's prescribed tire pressures shall be marked on the side of the vehicle or subframe near the tires. Unless otherwise specified (see 6.2), all marking shall be black on equipment painted forest green.

3.23 Vehicle weight classification. The contractor shall complete in the specified time (see 6.2) and submit the Government-furnished vehicle weight classification forms (see 6.4), one form covering each trailer chassis at GVWR or with the mounted equipment. The Government will inform the contractor of the resulting vehicle weight classification numbers. A variable

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vehicle classification kit, in accordance with MIL-S-40626, shall be furnished on each vehicle, with the trailer vehicle class displayed for GVWR or actual loaded equipment GVW, in a location approved by the contracting officer.

3.24 Toolbox. Each trailer chassis shall be equipped with a weatherproof toolbox on the equipment. The toolbox shall be metal or fiberglass and shall accommodate all cables, components, and tools required for maintenance and operation of the equipment. The lid shall be secured by a continuous hinge having a minimum 1/4-inch brass or stainless steel pin extending the full length of the lid. Two tension-type fasteners with strikers shall be provided to retain the lid in the closed position. A hasp shall be fastened to the lid to lock the toolbox. The hasp shall accommodate a padlock having a minimum 3/8-inch shackle. The fasteners, strikers, and hasp shall be attached to the toolbox by either bolts or rivets. When closed, the lid shall prevent rain and dirt from entering the toolbox. The lid shall open through on arc of at least 90 degrees and shall remain open by its own weight or by holding devices. At least 2-each drain holes with elastomer plugs shall be provided in the bottom of the toolbox.

3.25 Workmanship. Defective parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. All workmanship on the equipment shall be in accordance with the engineering and production standards of the heavy, farm and construction equipment trailer industry. The equipment shall withstand the operation specified herein without permanent deformation, breakage of connections, malfuncion, or component interference caused by incorrect workmanship. All parts of the equipment, before and after painting, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.25.1 Fabrication.

3.25.1.1 Aluminum and steel fabrication. Aluminum and steel shall provide original quality surface finish and shall be free from kinks and sharp bends. Aluminum and steel having eroded surface is not acceptable. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing, punching, and chipping shall be done uniformly, neatly, and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. The burned surfaces of flame-cut material shall be free of burrs, slag, and sharp edges. Precautions shall be taken to avoid overheating of metal. Heated metal shall be allowed to cool slowly, except where heat treatment is required. All bends shall be made with precise, unyielding dies or jig fixtures to insure uniformity of size and shape.

3.25.1.2 Fiberglass fabrication. Fiberglass reinforced plastic (FRP) shall be smooth surfaced and free from bubbles, glazing cracks, and discontinuities and shall have a uniform FRP thickness. All edges and openings of FRP shall be completely sealed to prevent water intrusion. The FRP shall be free of parting agents which impair paint adhesion. Welding, bonding, connections, and fabrication of the FRP shall be in accordance with the recommended methods of the manufacturer of the material for this application.

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3.25.2 Welders and welding.

3.25.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
An equal welder qualification program, subject to Government approval.

3.25.2.2 Metal welding. Welding shall be in accordance with the AWS and ASME codes, the American Welding Society's Welding Handbook, and as specified herein. Welds shall provide the maximum notch toughness and fatigue strength with a minimum of residual stresses. All welds made on high stress members shall be made from core or heavy-coated electrodes, gas shielded, or submerged arc. Electroslag, and electrogas processes, when available, shall be included for use on heavy sections, 1 inch or more in thickness. Fillet welds shall be provided, when necessary, to reduce stress concentrations. The surfaces to be welded shall be free from scale, paint, grease, and other foreign matter. When multiple layers of weld metal are required, each layer, or each layer after the third pass with multipass welding process, shall be throroughly cleaned before depositing another layer. All welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the parts or assembly. After welding, all weldments shall be free from slag (except mill scale), flux, weld spatter, and other factors which affect the quality of the weld. Except for vertical welding of the electroslag process, the work shall be positioned for flat welding whenever practicable. No undercuts or overlays will be permitted on any weldment. Where different metals are welded to each other, filler metal having strength of the lower strength materials shall be used. When the same metals, or metals having similar properties are used, the deposited filler metal shall develop the full design strength of the particular weld joint or weldment. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification. Weld joints exceeding 50,000 psi shall be inspected for soundness.

3.25.2.3 Aluminum welding. Aluminum welding shall be accomplished by the gas-metal-arc or the gas-tungsten-arc method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations. When machines are available, all aluminum welds shall be made by automatic machines. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed or chipped; hammered welds shall not be acceptable. Preheating for welding is permissible, providing the

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temperature does not exceed 400º F for a total time of 30 minutes. Aluminum weld processes requiring the use of welding flux shall not be used.

3.25.3 Brazing. Brazing shall be in accordance with ASM's Source Book on Brazing and Brazing Technology. Brazing of the aluminum and aluminum alloys shall be performed in accordance with the ASME code and other nationally recognized and accepted recommended procedures and standards of the aluminum manufacturing industry.

3.25.4 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Integral or separate washers or lockwashers shall be provided. Prevailing torque nuts are preferred in lieu of standard nuts and lockwashers. All capscrews, bolts, washers, and nuts shall be tight and correctly torqued. All fasteners and washers shall be finished with a corrosion preventative protective coating (including an eletrodeposited or mechanical plating or chemical conversion coating).

3.25.4.1 Critical bolted connections. Critical load-carrying bolted connections shall, in addition, be as specified herein. Boltholes shall be reamed as required. All critical junctures and joints shall be fastened with high-strength bolts with washers and nuts, hex or 12-point type. Bolts shall conform to IFI Standards 103 and 111, minimum SAE Grade 8, ASTM A449. Washers shall be of the flat hardened steel type, sized to spread the clamping loads such that brinnelling or embedding in the components members shall be prevented. Mating nuts shall be of the prevailing torque type in accordance with IFI Standards 100, 101, 107, and 108, minimum Class C, unflanged, or Class G flanged. Fasteners shall be lubricated prior to wrenching. To insure constant and correct torque performance, fastener installation accuracy shall be achieved by the turn-of-the-fastener method.

3.25.5 Riveted connections. Riveted connections shall be in accordance with SAE J492. Rivet holes shall be accurately punched or drilled and shall have the burr removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.25.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use. Iron castings shall be in accordance with the Iron Castings Handbook.

3.25.7 Forgings. Forged components shall be in accordance with the Forging Industry Association Handbook.

3.25.8 Machine work. All machined parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of these parts.

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3.25.9 Cleanliness of hydraulic and pneumatic systems. The hydraulic and pneumatic systems shall be purged to remove any foreign material after the appropriate assembly phase is completed. The sequence of purging shall be first through the lines and then second through both the lines and the components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection (size 9 MD). Failure of any size 9 MD trailer shall be cause for disassembly, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembly part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article trailer chassis. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components (size 9 MD). Parts and components detailed on the size 9 MD trailer drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.10 and the size 9 specification sheet.)

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Comparison inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection of packaging (see 4.7).

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4.3 First article inspection.

4.3.1 Examination. The first article trailer chassis shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection when defect(s) involve components not covered by drawings, or for performing the disassembly inspection specified in 4.1.2 when defects(s) involve components covered by drawings.

4.3.2 Tests. The first article trailer chassis shall be tested as specified in 4.6.2.1 through 4.6.2.7. Failure of any test shall be cause for rejection when the failure(s) involve components not covered by drawings, or for performing the disassembly inspection specified in 4.1.2, when the failure(s) involve components covered by drawings.

4.3.3 Packaging. Packaging of the first article trailer chassis shall be examined as specified in 4.8.1.3. Presence of one or more defects shall be cause for rejection.

4.4 Comparison inspection. The government may select trailer chassis at any time during the contract production period and subject these trailer chassis to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on trailer chassis selected at random from those which have been accepted by the Government and will not include the previously inspected first article trailer chassis. In addition to any test specified as part of the comparison inspection, the Government reserves the right to conduct any and all other tests contained in this specification as part of the comparison inspection and failure of such additional tests shall gave the same effect as failure of those tests specified as comparison inspection.

4.4.1 Inspection failure. Failure of a comparison inspection trailer chassis to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the comparison inspection trailer chassis and shall be cause for refusal by the Government to continue acceptance of production trailer chassis until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trailer chassis previously accepted and produced under the contract. Any deficiencies found as a result of the comparison inspection will be considered prima facie evidence that all trailer chassis accepted prior to the completion of comparison inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Quality conformance inspection shall be conducted on production trailer chassis. Sampling for inspection and accept/reject decision shall be in accordance with MIL-STD-105.

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4.5.2 Examination.

4.5.2.1 Samples. Samples selected in accordance with 4.5.1, shall be examined for the major and minor defects specified in 4.6.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.6 Inspection procedure.

4.6.1 Examination. The trailer chassis shall be examined for the defects listed in table V.

TABLE V. Examination schedule (requirements).

Number	Characteristics	Requirement Paragraph
Major		
101.	Material or common parts not as specified.	3.3, 3.4
102.	Design criteria not furnished, incorrect, or not as specified.	3.4.1
103.	Maintainability not as specified.	3.5
104.	Transportability features incompatible.	3.6 thru 3.6.5.2
105.	Human factors characteristic not as specified.	3.8
106.	Safety characteristics not as specified.	3.9
107.	Environmental features not furnished or not as specified.	3.10
108.	Equipment chassis interface components not as specified.	3.11
109.	Specifications of trailer chassis not as specified.	3.12
110.	Trailer body option not as specified.	3.13
111.	Chassis component option(s) not as specified.	3.14
112.	Military vehicle electrical system option not as specified.	3.15
113.	Electromagnetic interference characteristics not as specified.	3.17
114.	Lube and lubrication features not as specified.	3.18
115.	Servicing and adjustments not made, incomplete or not as specified.	3.19
116.	Predelivery check list not followed as specified.	3.19
117.	Finish treatment and painting missing, incorrect, or not as specified.	3.20
118.	Plates, marking, and weight class missing, incorrect, or not as specified.	3.21 thru 3.23

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TABLE V. Examination schedule (requirements). (Cont'd)

Number	Characteristics	Requirement Paragraph
119.	Toolbox not as specified.	3.24
120.	Workmanship not as specified.	3.25 thru 3.25.9
121.	Leaks in oil, coolant, fuel, intake air, bearings, hydraulic, air brakes and other lines.	Section 3
122.	Seals do not prevent entrance of foreign material nor prevent leakage of fluids, greases, or air.	Section 3
123.	Assembly incorrect or not complete.	Section 3
124.	Parts or components missing or do not function.	Section 3
Minor		
201.	Cleaning not as specified.	Section 3
202.	Departure of workmanship from commercial pratice.	Section 3

4.6.1.1 Certification. The contractor shall insure that the components and materials used in the trailer chassis are manufactured, examined, and tested in accordance with the requirements and references specified herein. As a minimum, the contractor and his component sub-suppliers shall be required to satisfy the criteria for the following:

a. In effect at actual time of manufacture:

1. DOT FMVSS.
2. DOT FMCSR.
3. California Vehicle Code.
4. DOL OSHA Standards and Regulations.

b. In effect at time of solicitation of bid:

1. Industry Standardization Document Compliance.
2. Comply with the performance requirements specified.
3. Meet the physical specifications specified.

4.6.2 Tests.

4.6.2.1 Test procedures. Prior to testing, lubricate the trailer chassis with lubricants as specified in 3.18. Lubricants shall be those designated for use in the ambient temperature at which the test is being performed. Before any road and off-road terrain tests are conducted, measure longitudinal and diagonal points and critical mounting points on the trailer chassis. After tests are completed, check the measurements of these points and visually inspect the trailer,

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including all welds and fasteners. During the tests, all trailer chassis service lights, and if applicable, the blackout lights, shall be illuminated, turn signals operated on turns, and hazard warning operated during stops. The trailer chassis GVWR and payload shall be as specified in table I. The apparatus minimum standard of accuracy shall not exceed the following:

Type of measurement	Percent of measured value
a. Deceleration.	± 2
b. Force.	± 2
c. Linear dimension.	± 1
d. Speed.	± 2
e. Stress	± 2.5
f. Weight	± 2

4.6.2.2 Stress analysis. Except for size 9, a stress analysis for the trailer chassis shall be calculated and furnished, containing load, shear, and moment diagrams (see 3.4.1) both while blocked up for equipment operation and while on its running gear. Compare the analysis with the maximum allowable stress. Demonstrate the ability of the frame to withstand, without permanent deformation, the operating stress, while jacked-up on stands, and the transported stress, while being towed on the running gear. Begin by determining the payload, that amount of weight allowance for the operating equipment and accessories and the correct payload distribution, which is necessary for the trailer chassis to weigh-in at a value equal to the maximum chassis GVWR cited on the manufacturer's VIN name plate. Further, determine the test weight necessary to achieve 1.75 time this payload. The test weights shall be imposed on the 3-each operating equipment skid mounting crossmembers (see 3.11) or, in the case of size 32 and others with a flat platform or special bodies, imposed over the most critical 8-linear-feet of the bed area. Support the trailer chassis off of the suspension and tires. At first, support the trailer chassis only on the drawbar front tow eye and on the points where the suspension attaches to the chassis frame. Prior to testing, preload the chassis frame to 1.75 time the specified payload to relieve stresses, then unload. Establish and record the dimensional relationship of trailer chassis points at the lowest outside edges of the mainframe members, below the load bed, on both the roadside and the curbside as follows:

Point A – Frame height at drawbar-to-bed junction (normally, No. 1 crossmember).
Point B – Frame height just ahead of end of frame.
Point C – Frame height midway between points A and B.

Impose a test load of 1.75 times the specified payload and record all 6 points. Unload and record all 6 points again. Compare the first set of record point measurements with the third set of recorded measurments. Repeat this test with the trailer chassis only supported on the toweye and the rear operating equipment leveling leg mount points to the chassis frame. If no permanent deformation on any member of the trailer chassis has occured, the indicated trailer chassis payload rating shall be acceptable.

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4.6.2.3 Road runs. The following road runs shall be performed: Couple the trailer chassis to the truck and tractor prime movers and tow as specified for a total distance of at least 400 miles for the trailer chassis at the variable speeds and under the conditions indicated. The load shall be the payloads necessary to achieve GVWR.

Course	Mileage	Speed (variable)	Loading
Primary highway	250	Up to 55 mph	Loaded
Unpaved roads	100	Up to 30 mph	Loaded
Off-road terrain	50	Up to 20 mph	Loaded

During the road runs and while towing rated payload, perform the following and record actual results:

a. Tow the trailer chassis on side slopes up to 10 percent, having the curbside downhill and having the roadside downhill. Make full throttle starts and panic stops on the side slope.
b. Tow the trailer chassis over obstacles having the height as specified in the specification sheet such that all combinations of the wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and, after returning the suspension to normal, measure the diagonals and parallels for evidence of misalinement. Questionable tire-to-surface contact shall be retested with single sheets of newspaper under the tires. Ability to remove the newspaper without tearing from under any tire shall be evidence of insufficient tire-to-surface contact.
c. For determining multiple axle equalization, weigh the weight imposed on each trailer chassis axle (each axle on a seperate scale) simultaneously and record the weight on each axle. Jack up each axle separately and return it to its scale, again recording the weight imposed on both the raised axle and those not raised. After each axle has been raised and all axle weights recorded, compare the suspension equalization differences between axles.
d. Tow the trailer chassis over a chatter course until at least 2600 either upward or downward accelerations of magnitudes indicated herein have been applied to a rear wheel center. The course shall be of such construction that the acceleration magnitudes indicated below can be obtained by adjusting the vehicle speed without the frequency of the accelerations exceeding 15 cycles per second (cps) or becoming less than 3 cps. The maximum acceleration magnitude indicated below shall not be exceeded, and accelerations having magnitudes less than the minumum shall be disregarded. The course shall be of such length that a wheel center receives, in one pass over the course, at least 50 consecutive vertical acceleration cycles of any magnitude less than the maximum specification herein. The accelerations shall be measured by a trial test to establish the required vehicle test speed. The recording instruments shall then be removed or disconnected and the test completed at established speed. The acceleration values used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the vehicle test speed is

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held constant throughout the test shall be provided. Accelerations shall be measured on a running-gear member, usually on axle or spindle, that has a motion identical to that of the wheel center. The magnitude of the accelerations shall be not more than 6 g nor less than 3 g, where 9 represents the weight imposed.

e. Tow the trailer chassis through mud at least 6 inches deep for not less than 100 feet. Any prime mover may be used for this phase.
f. Tow the trailer chassis over highway v-ditches (standard FHWA Bureau of Public Roads type) at least 36 inches wide and 18 inches deep. Negotiate the ditches both perpendicularly and at an angle of 45 degrees to the direction of travel for no fewer than 10 times. Inspect the underside for damage.
g. Measure and record the DOT FMCSR deviation or off-tracking of trailer chassis with respect to the prime mover when the trailer chassis is towed straight ahead on a paved surface.
h. Jackknife the trailer chassis at angles up to 60 degrees for all CCE sizes and 40 degress for size 9 MD trailer chassis from straight ahead both to the left and to the right.
i. Turn the prime mover, towing the trailer chassis, in a rolling sharpest-turning circle for two complete turns and repeat in the opposite direction.
j. When uncoupled from the truck, raise the front of the trailer chassis not less than 4 inches from a concrete surface and rapidly allow the trailer chassis to fall on the support leg or landing jack. Visually inspect and compare before and after measurements for evidence of permanent deformation.
k. For size 9 ND trailer chassis only, maneuver the uncoupled trailer chassis on a paved surface for not less than 50 feet in distance and complete not less than one full 360-degree rolling turn.

4.6.2.4 Brakes. Test the trailer chassis for conformance to the DOT Motor Carrier Safety Regulations and Federal Motor Vehicle Safety Standards, the California Vehicle Code, and to the requirements specified herein. Runs shall be conducted with rated payload unless different loading is specified. In addition to the DOT and California legal tests, perform the following:

a. With the trailer chassis stopped on a paved surface, uncouple the truck brake lines from the trailer chassis and bleed down all trailer chassis air reservoirs. Determine by the drawbar pull method or on actual grades the maximum holding ability of the trailer chassis emergency parking brakes. Release the emergency parking brakes. Release the emergency parking brakes using only tools supplied with each trailer chassis or with an external resupply of compressed air; simulate earthmover equipment towing without air supply.
b. Test the trailer chassis airbrake system as specified in TTMA RP No. 12 and RP No. 44.
c. Remove the chock blocks and place them, first to prevent the trailer chassis from rolling forward, and then repeat to prevent the trailer chassis from rolling rearward. Do not unsnap the captive chain from the cradle loop.

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4.6.2.5 Fording. Tow the trailer chassis through creekbed water crossings not less than the depth specified for not less than 2 minutes. Stop the trailer chassis in the water and turn off the lights. Turn on the lights and continue towing for a total time of not less than 5 minutes. Park the trailer chassis out of the water and inspect the trailer chassis.

4.6.2.6 Failure criteria. Inability of the trailer chassis to complete any of the tests specified in 4.6.2.2 through 4.6.2.5, or nonconformance to any requirement specified herein, shall constitute failure of that test.

4.6.2.7 Lifting and tiedown. With the trailer chassis anchored by means other than the lifting or the tiedown attachment or locations being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the attachments. Nonconformance to 3.6.5.1 or 3.6.5.2, weld failure, permanent deformation of the trailer chassis or its tiedown provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6.2.8 Railroad car impact. When specified (see 6.2), anchor the unpackaged trailer chassis on a domestic service railway freight car (blocking is permitted). Hump the car, with the trailer chassis tied down, at speeds of 4, 6, and 8 mph, (+5 –0, percent mph) respectively, against 5 empty freight cars with their brakes set and draft gear extended. The trailer chassis shall face in the direction of the 5 cars. Repeat the procedure at not less than 8 mph with the rear of the trailer chassis in the direction of the 5 cars. The trailer chassis may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the trailer chassis in the proper direction on the freight car. Once the test has begun, there shall be no readjustment of load nor any reconditioning of the blocks, chocking material, or tiedowns except as necessary to reverse the direction of the trailer chassis. After each impact, the trailer chassis shall be inspected. After completion of the railroad impact test, an operational checkout of the trailer chassis shall be conducted. Nonconformance to 3.6.2 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

125. Trailer chassis not preserved as specified (see 5.1.1.1).
126. Repair parts not preserved as specified (see 5.1.1.2).
127. Maintenance tools not preserved as specified (see 5.1.1.3).
128. Consolidation not as specified (see 5.1.1.4).
129. Strapping not as specified for level A (see 5.1.1.4).
130. Marking illegible, incomplete, or incorrect (see 5.3).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Trailer chassis. The trailer chassis shall be preserved in accordance with the level A (mobile) requirements of MIL-STD-281.

5.1.1.2 Repair parts. Repair parts shall be preserved in accordance with MIL-P-116.

5.1.1.3 Maintenance tools. The tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.1.1.4 Consolidation. The repair parts and maintenance tools shall be placed in boxes conforming to PPP-B-601, Overseas Type, Grade B. Strapping shall conform to QQ-S-781, Class 1, Type 1 or IV, size as applicable; and, unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2 Packing. Packing shall be level A, or commercial, as specified (see 6.2).

5.2.1 Level A. The trailer chassis shall be prepared for mobile shipment, and consolidated boxes shall be placed on the trailer chassis and secured by strapping, bolting, or similar means. Strapping shall be as specified in 5.1.1.4.

5.2.2 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Level A packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

5.4 Depreservation guide. Two depreservation guides (see 6.13) shall be prepared. One guide shall be placed in a waterproof envelope marked “Depreservation Guide” and secured in a conspicuous location on the van body. The other guide shall be placed in the package with the technical manuals and the package marked to indicate the depreservation guide is inside. Unless otherwise specified (see 6.2), DA Form 2258, “Depreservation Guide for Vehicles and Equipment” shall be used.

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6. NOTES

6.1 Intended use. The trailer chassis is intended for the mounting and transporting of various special items of equipment which require a low overall height over highway and cross-country terrain. The intended equipment items include the following:

Water Purification Unit, 600 GPH	MIL-W-52116, Type II
Distillation Unit, Water, 150 GPH	MIL-D-52386
Lubricating and Servicing Unit	MIL-L-10269
Welding Shop, Trailer Mounted	MIL-W-52584
Radiographic Unit	
Air Compresser sets w/Tools.	MIL-C-555
Hydraulic Service and Repair Units.	
Outside Light Sets.	
Power Units (Eng.-Gen. Sets).	
Environmental Control Unit.	
(Air Conditioners, Heat Pumps, etc.)	
Maintenance Shop Sets	
Woodworking Shop Set	MIL-S-52339
Pioneer Tool Shop Set	
Diesel Engine Power Units	
(Mech. Drive for Rock Crushers, etc).	

The trailer chassis are intended to be towed by military design trucks, commercial trucks, and self-propelled military equipment such as crawler and wheeled tractors, scoop loaders, cranes, and motor graders.

6.2 Ordering data. Acquisition documents should specify the following administrative ordering data and technical ordering data:

a. Title, number and date of this specification and of the applicable size specification sheet(s).
b. Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1.).
c. Time frame required for submission of the first article plant model (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. When the SSPC treatment and painting method is acceptable in lieu of the MIL-T-704 method (see 3.20).
f. Cite the appropriate service and what marking is required (see 3.22).
g. Cite color of marking on forest green equipment if different from black (see 3.22).
h. What time frame is required for the contract to complete the equipment physical characteristics and submit this technical data to the Government for calculation of the vehicle weight classification number (see 3.23).

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i. When railroad hump testing of the trailer is required to verify the tiedown attachments and trailer assembly (see 4.6.2.8).
j. Disposition of the first article models, if required to be different than as specified, remain as a model for all other production trailers (see 3.2).
k. Degree of preservation and packing required (see 5.1 and 5.2).
l. When strapping shall be other than as specified (see 5.1.1.4 and 5.2.1).
m. When a depreservation guide form, other than DA Form 2258, is required (see 5.4).
n. When and as specified the technical characteristies and specific details listed as follows in the Technical Component ordering data list are required.

Technical Component Order Data List. The technical features shall be selected and required to be furnished with each trailer chassis or shall not be needed.

T-1	Furnish optionally required features and components as specified, including size XX in 1.2 (see 3.1.1 and specification sheets).
T-2	Furnish the size 9 military design trailer chassis with the optionally required features and components as specified on specification sheet MIL-C-52437/6 and as follows (see 3.1.2 and MIL-C-52437/6): - Other than the Dash-l quarter elliptic trailing arm springs. Specify payload and the spring dash No. on the leaf spring drawing (Ref. D13212E3632). - Specify when fenders are required to be furnished (not furnished with cargo body installation). - Specify when a front platform per drawings is needed. - Specify when the rear commercial jack stands are required to be furnished. - Specify any other chassis characteristics required.
T-3	Furnish the size and quantity of running gear for flotation, as specified (see 3.1.3 and specification sheets).
T-4	Load and hump the trailer chassis and its equipment payload (or at GVWR) on a railroad car (See 3.6.2 and 4.6.2.8).
T-5	Provide the trailer chassis with its equipment payload capable of being loaded and transported within C-130 or other aircraft (see 3.6.4).
T-6	Provide trailer chassis compatible with the vehicles specified in addition to or in lieu of these cited [see 3.7.1 and 3.9 (k) and specification sheets]. Specify vehicles. Specify a slow-moving vehicle emblem on trailer chassis for use when trailer is towed by off-road equipment.

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T-7	Specify the intended operating equipment or set payload, if known, or cite the GVWR for the applicable size specification sheet trailer chassis. Also, specify which equipment mount boltholes on the trailer chassis shall be drilled by the contractor. Also, specify when and how to mount and secure the intended equipment permanently onto each trailer chassis (see 3.11).
T-8	Furnish brake systems as specified to coincide with all of the Government user's brake system needs specified (see 3.12, table II, VMRS 13).
T-9	Furnish air ride trailer suspension system to protect the operating equipment from damaging ride (see 3.12, table II, VMRS 16-008).
T-10	Furnish a hubodometer assembly, when needed by Government users (see 3.12, table II, VMRS 51-002-001).
T-11	Furnish, a trailer frame of special low temperature steel for cold region (arctic) operations (see 3.12, table II, VMRS 77-012-002).
T-12	Furnish a trailer chassis with a trailer body as specified in table III or as detailed by the Government user (see 3.13 table III).
	T-12A. Specify wood planking or aluminum tread plate floor assembly. T-12B. Specify when other than the contractors selection of FRP or aluminum body panels. T-12C. Specify when and which logistic tracks are required, type E or type F. T-12D. Specify when, where, and the type of doors, awnings, or hinged panels required. T-12E. Specify when an inside body lining is required. T-12F. Specify when an insulation between the outer skin and inner lining is required.
T-13	Furnish various chassis components as specified (see 3.14).
T-14	Furnish a military vehicle electrical system, in addition to the 12 volt TTMA commercial system, to permit towing of the trailer chassis by Military Design trucks (3.15).

6.3 First article model. Any changes or deviations of production chassis from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish trailers conforming to this specification.

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6.4 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the trailer chassis. The contractor should then be requested by the contracting officer to apply the weight classification number(s) to the trailer chassis. When operating equipment and its weight is known or can be estimated, these data shall be used for determining the vehicle classification. Otherwise, the GVWR of the trailer chassis shall be used for the vehicle classification (see 3.23).

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each trailer chassis.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of trailer chassis.

6.7 Registration numbers. The contracting officer should furnish a registration numbers to be applied to the trailer chassis (see 3.22).

6.8 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the trailer chassis, covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91 (see 3.18).

6.9 Supplementary quality assurance provisions (SQAP). The contracting officer should require the contractor to maintain records on size 9 MD trailers of all SQAP inspections. A suggested paragraph is as follows:

"For the military design trailer chassis of MIL-C-52437/6, Size 9, the contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken, when applicable."

6.10 Definition supplementary quality assurance provisions (SAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure inter-changeability of repair parts.

6.11 Preproduction pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the trailers in accordance with this specification.

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6.12 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.4).

6.13 Depreservation Guide. The contracting officer should arrange to furnish DA Form 2258, the Depreservation Guide, when requested by the contractor (see 5.4 and 6.2).

6.14 Technical Information Package. The contractor should complete and submit a Technical Information Package, depicting the trailer and its components as it will be furnished. The Technical Information Package should be submitted with all bids and proposals.

Custodians:
Army - ME
Air Force - 99

Preparing activity:
Army - ME

Review activity:
Army - AT
Air Force - 84

Project 2330-0786

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